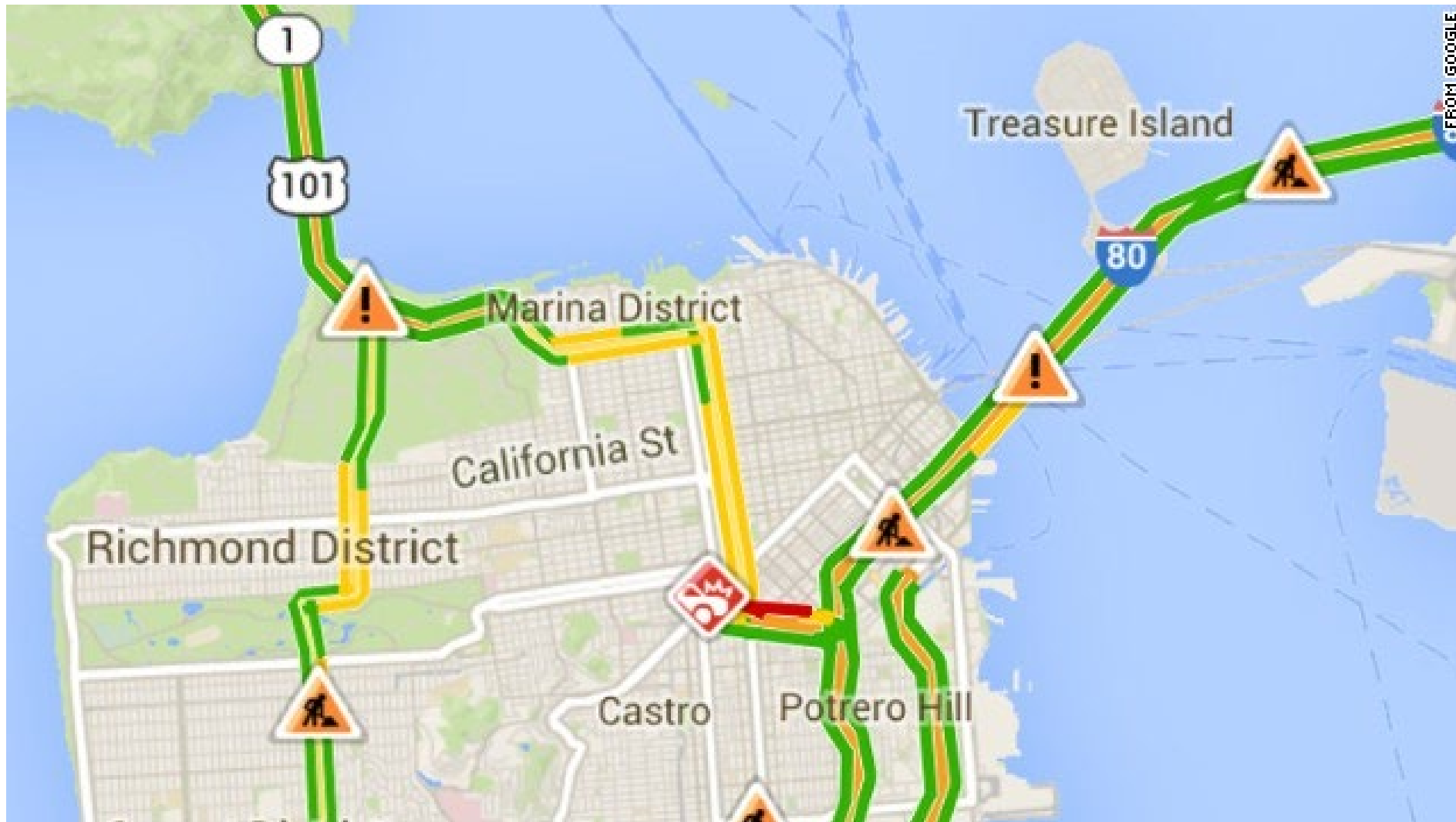


Data Visualization: The Basics

Tim Dunn, Visualization Specialist in Research Computing and the Center for Research Data and Digital Scholarship
Melissa Cantrell, Scholarly Communication Librarian in the Center for Research Data and Digital Scholarship



You know more than you think you know





Agenda

You will leave today with:

- 1. How do data visualizations work?
- 1. Categorization of data visualization
- 1. Elements of design
- 1. Common types of visualization
- 1. Best practices





How do data visualizations work?

Building the foundation



WHAT?

Data visualization is a **graphical display of information** for two main purposes:

- sense-making (e.g. analysis)
- communication.

WHY?

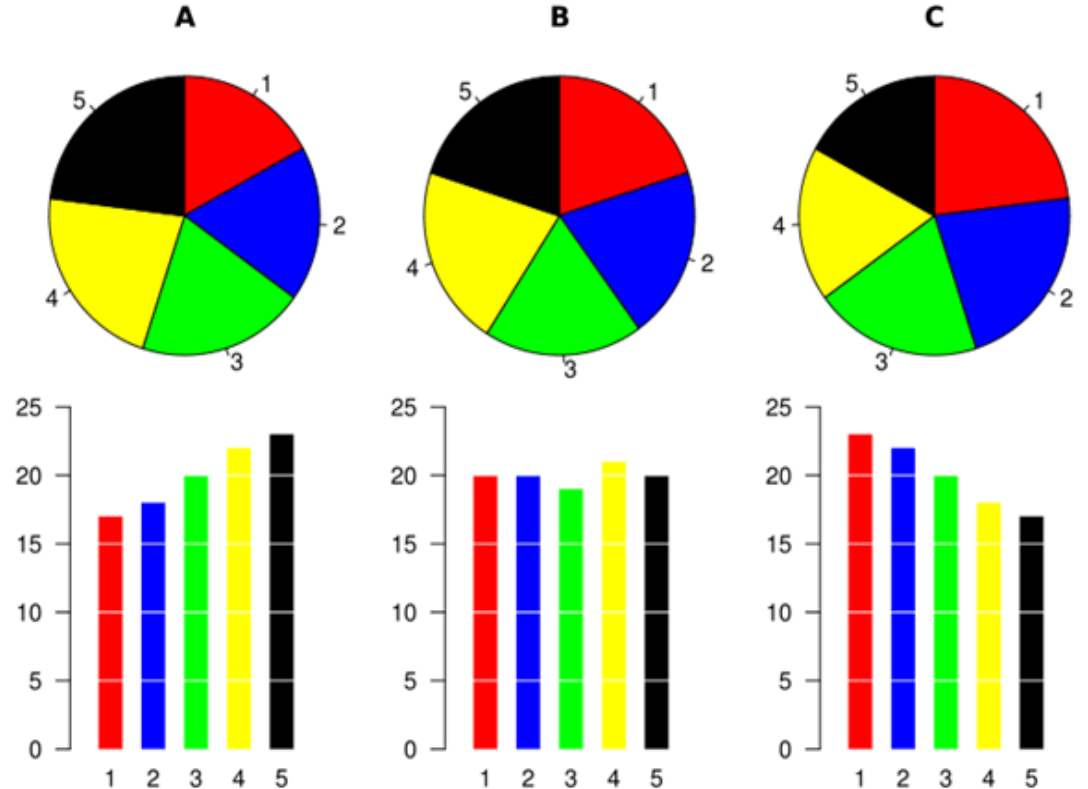
Visual displays **give form to information** which before had none. Human visual perception and cognition is particularly strong.

<https://www.interaction-design.org/literature/book/the-encyclopedia-of-human-computer-interaction-2nd-ed/data-visualization-for-human-perception>

Constructing meaning with data visualization

We use data viz to **demonstrate patterns**.

...But we cannot forget that displaying patterns involves careful choices on our part.

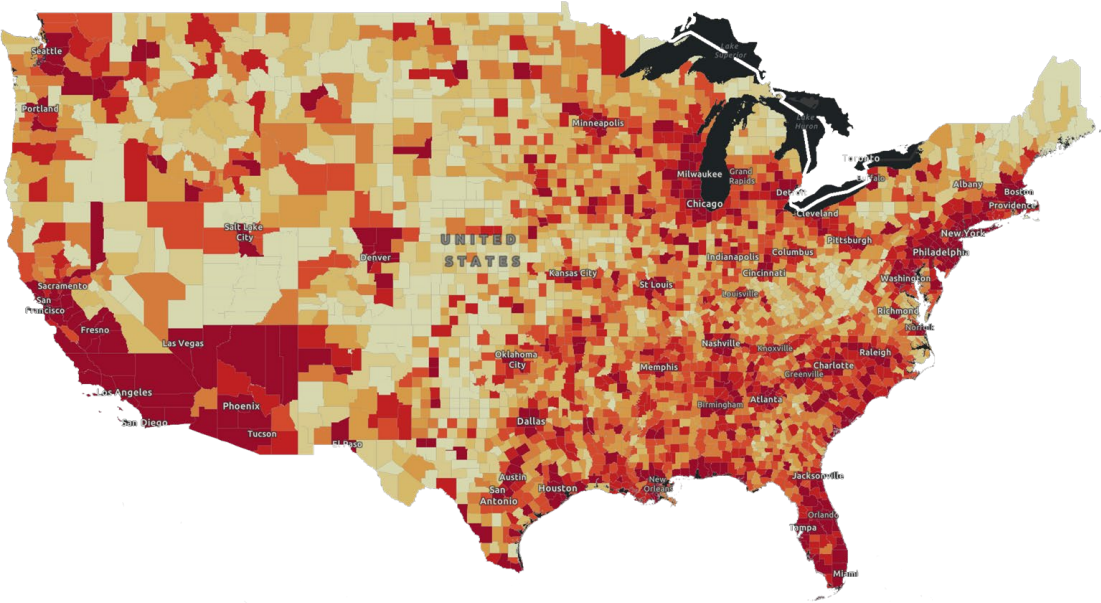


https://eazybi.com/blog/data_visualization_and_chart_types/

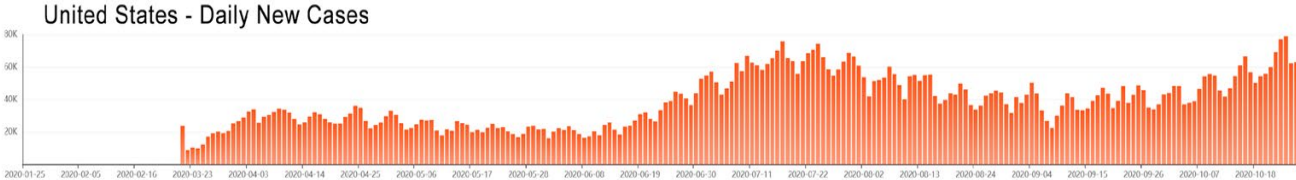




Constructing meaning with data visualization



All three of these plots show plots of COVID-19 data from the same day. (Mon Oct 26)

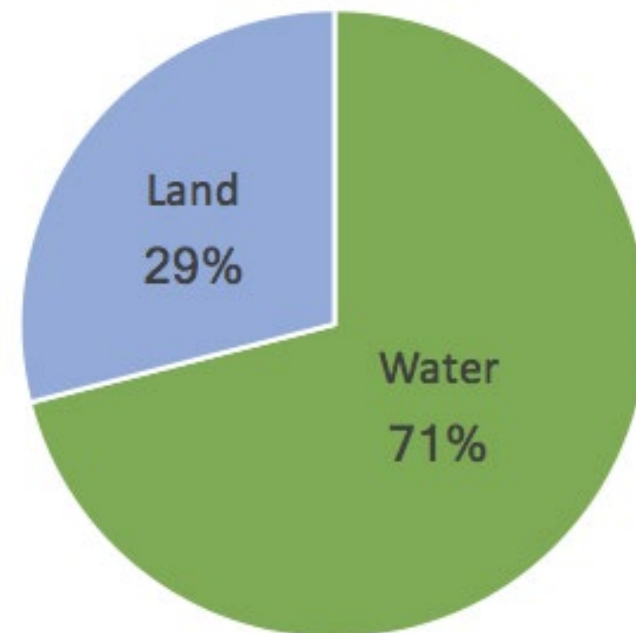




Constructing meaning with data visualization

We use both our **conscious** and **subconscious** (“snap”) judgements to quickly interpret visual cues

Land to Water Ratio on Planet Earth



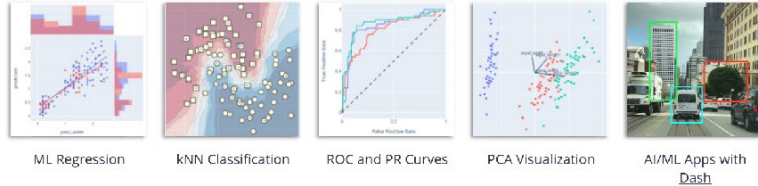


Catagorization of data Visualization

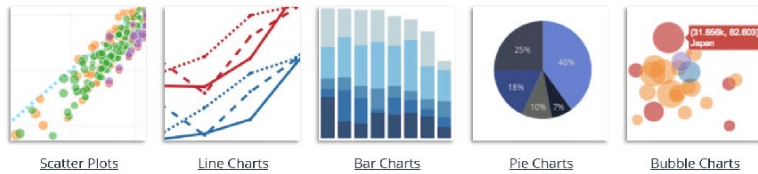


2D Plots, charts, graphs

Artificial Intelligence and Machine Learning



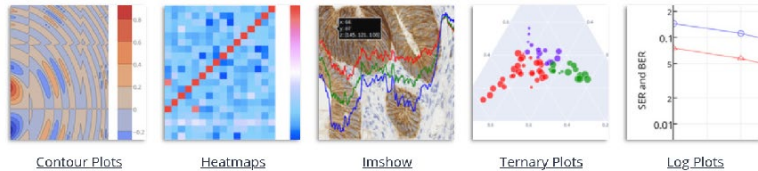
Basic Charts



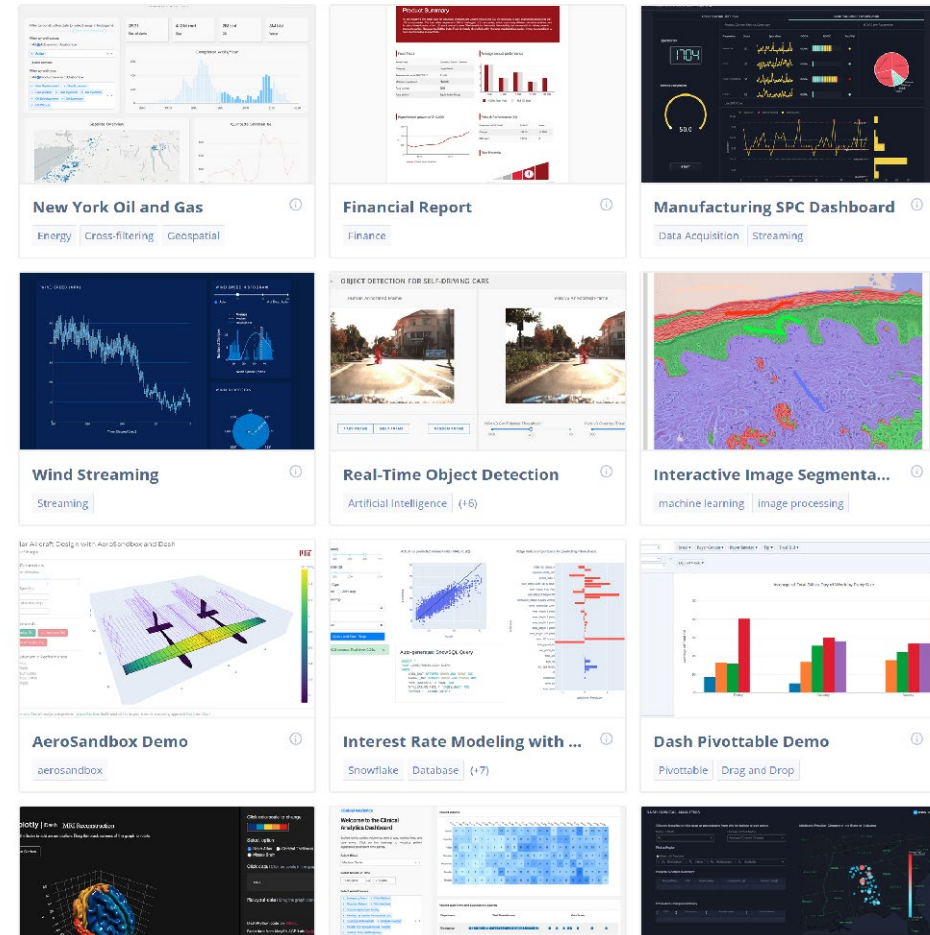
Statistical Charts



Scientific Charts

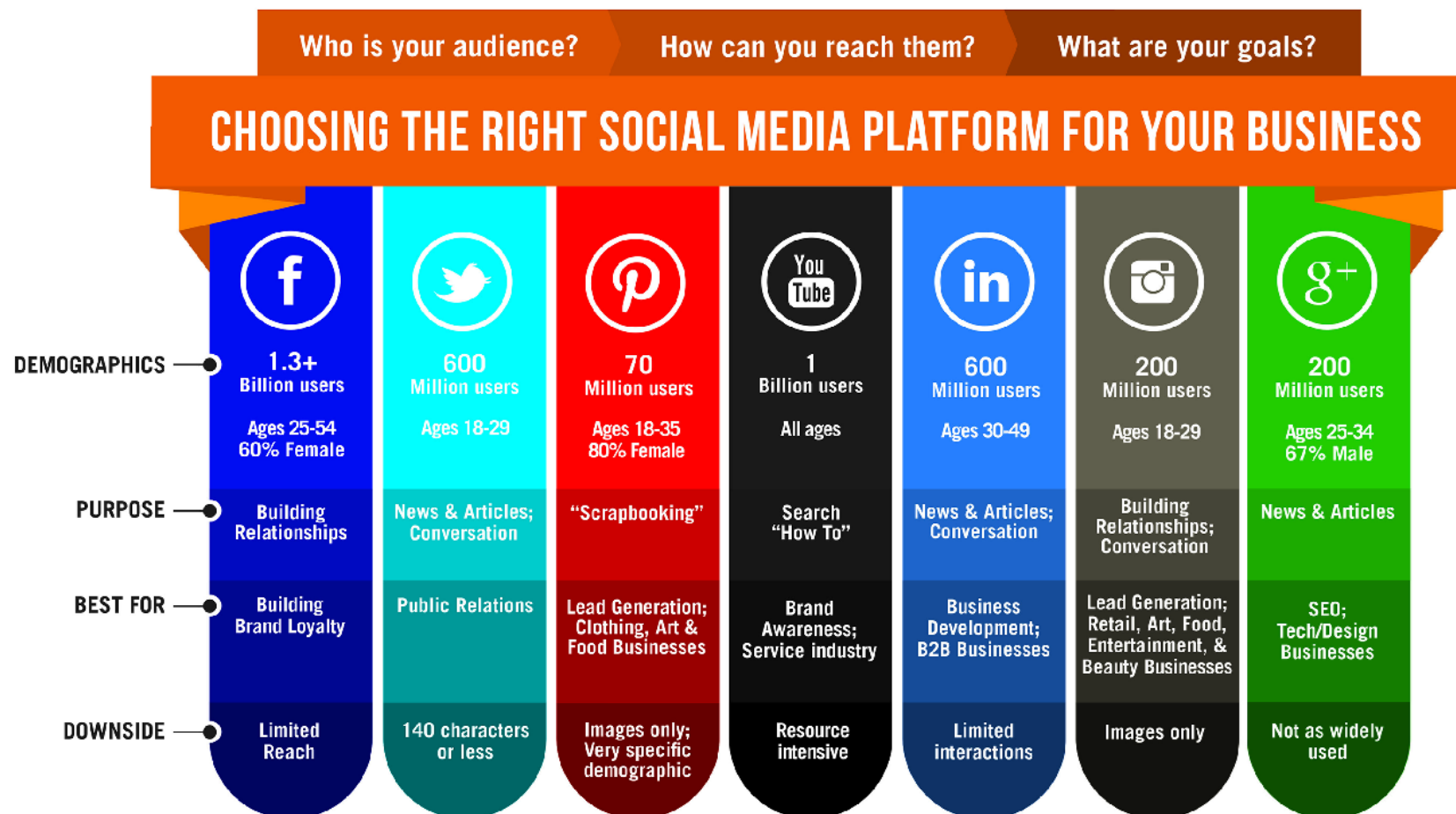


<https://plotly.com/python/>



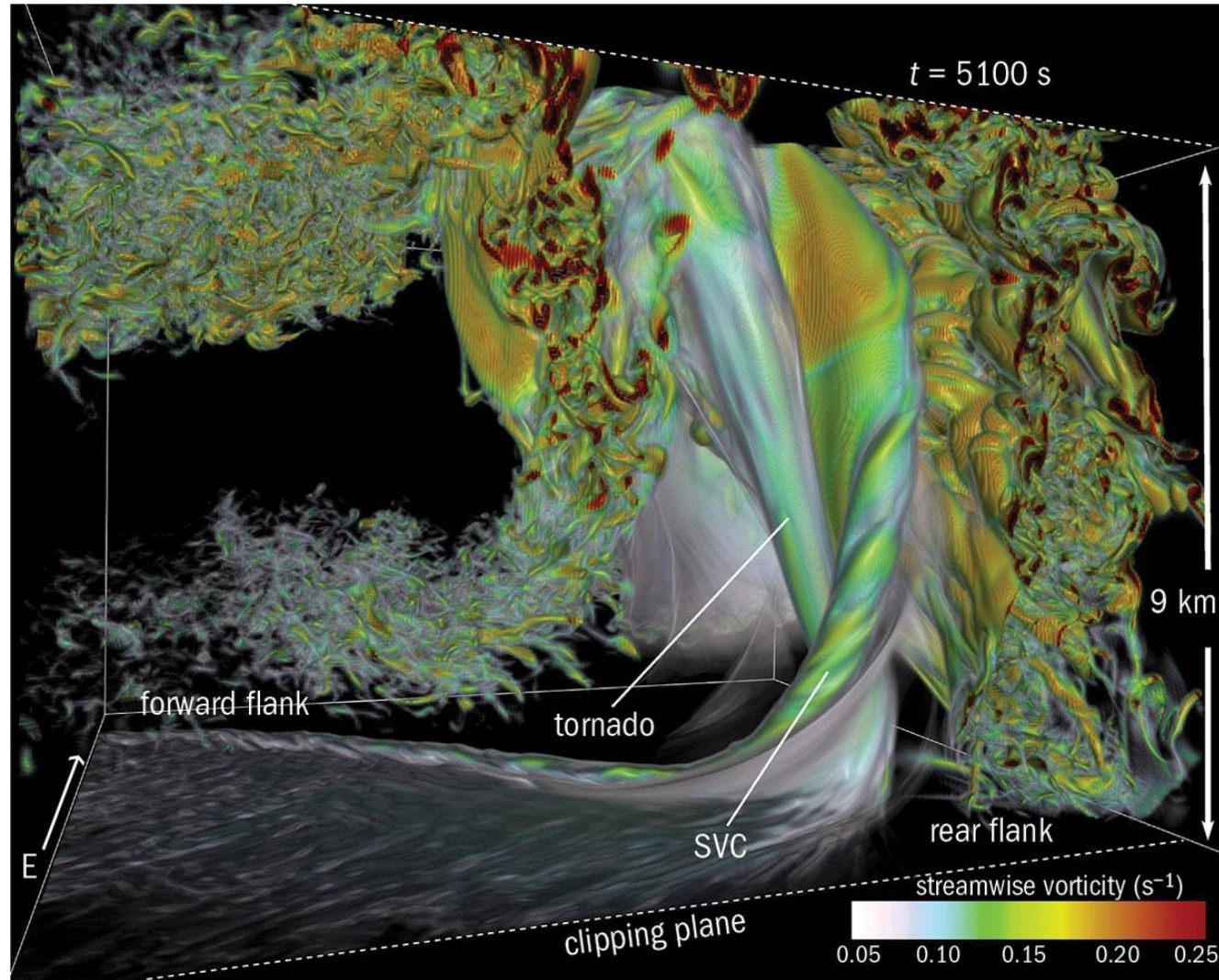
<https://dash-gallery.plotly.host/Portal/>

Infographics / Illustration



<https://us.accion.org/resource/choosing-right-social-media-platform-your-business/>

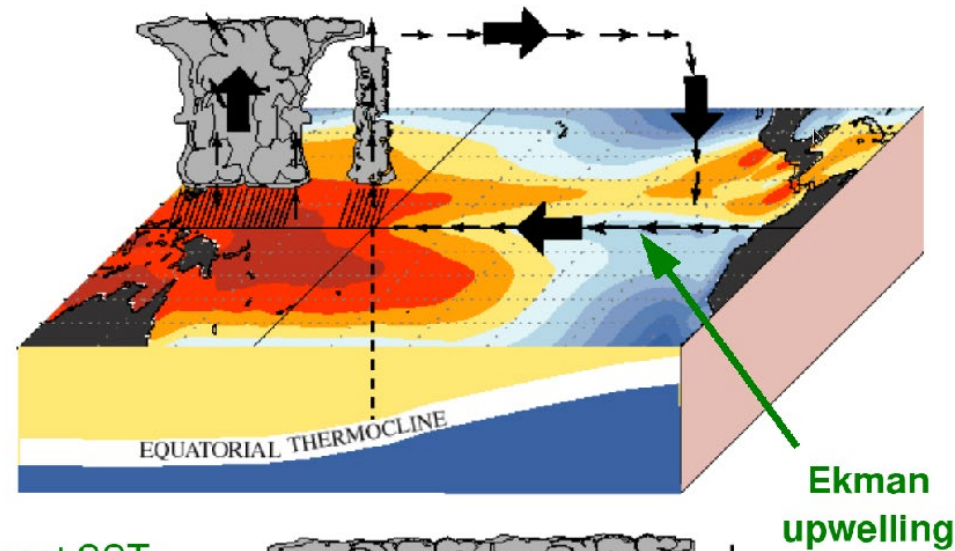
3D Visualization



Leigh Orf – Tornadogenesis simulation visualized with Vapor2

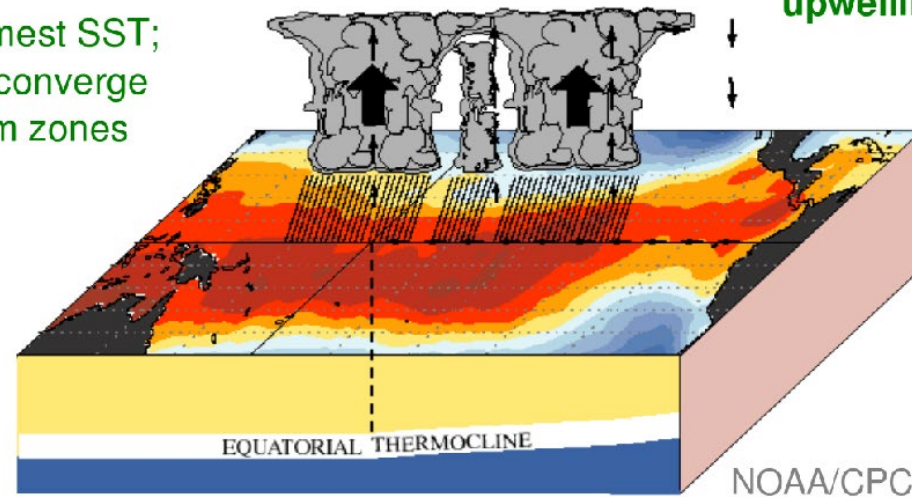
Composite

Normal



rain follows warmest SST;
surface winds converge
onto rainy/warm zones

El Niño



<https://www.gfdl.noaa.gov/el-nino-media/>



Media used for data visualization

Data visualization can be used in various different media for different expressions of your story.

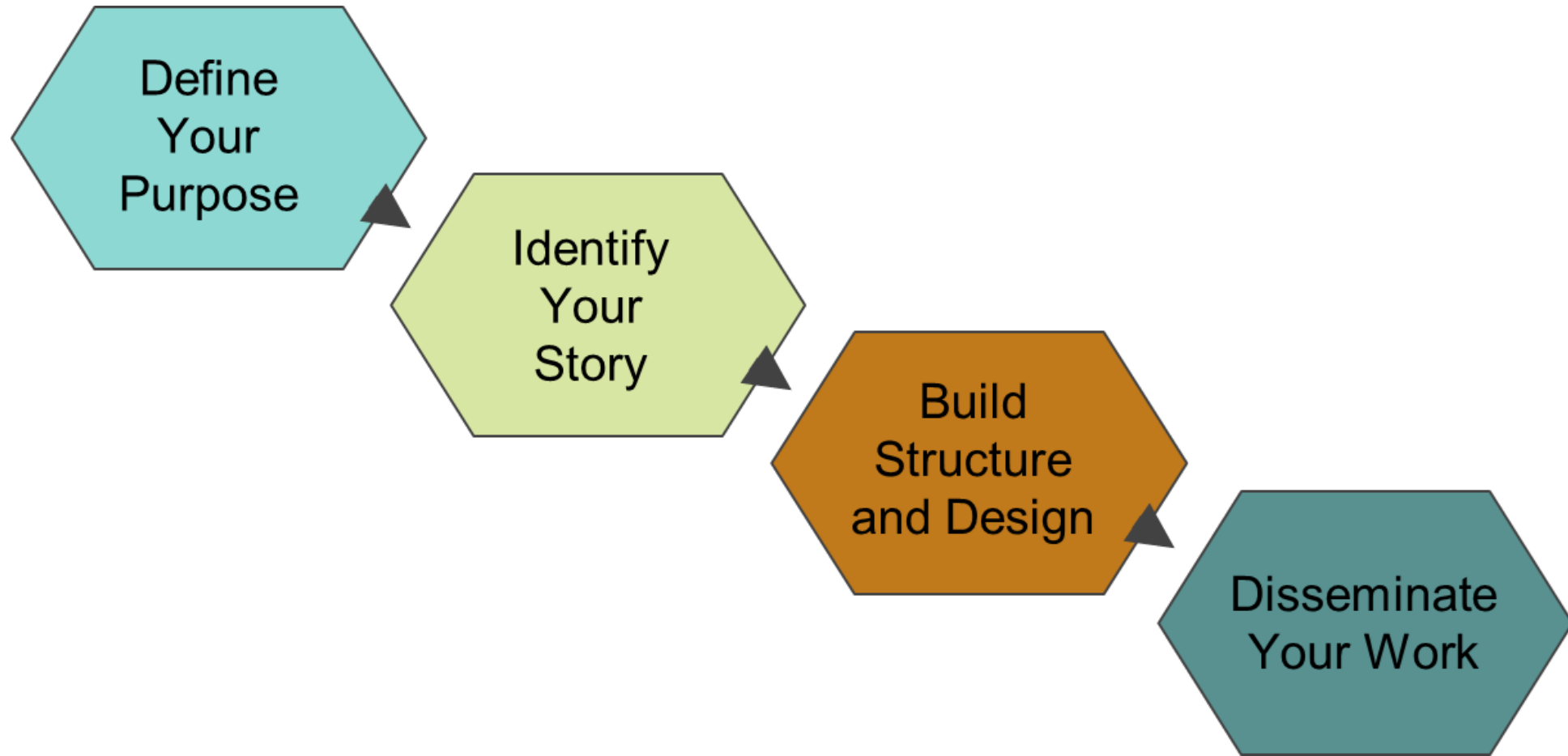
- Print
- Social Media
- Static web pages
- Interactive web pages
- Animations/movies
- Virtual (eg VR, AR, XR)





Elements of Design

Elements of Design in Effective Data Visualization



Elements of Design in Effective Data Visualization



Purpose: Start from a motivated position.

Consider your audience, what they care about, their data literacy, and what they need to make decisions.

<http://www.mediafactory.org.au/2015-media1-projects-broadcast-vs-post-broadcast/files/2015/05/audience-qg9z4y.jpg>



Elements of Design in Effective Data Visualization



Purpose: Start from a motivated position.

How should the audience receive or interact with the data?

<https://www.flickr.com/photos/tringa/23861635520>



Elements of Design in Effective Data Visualization

Story: What's Your Point?

Consider an Exploratory versus an Explanatory approach.

Ask yourself, do you want your audience to develop their own conclusions or do you want to convey a specific message?



https://pixabay.com/p-89198/?no_redirect

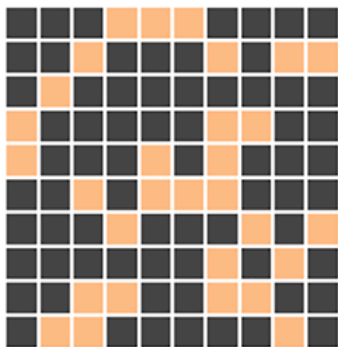
Ebola spreads slower, kills more than other diseases

This simulation shows how quickly 10 diseases, from more fatal to less fatal, could spread from one person to 100 unvaccinated people.

■ Newly infected ■ Recovered ■ Deaths Day 78

RUN NEW SIMULATION

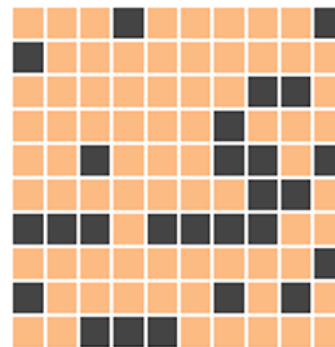
Ebola ⓘ
DAYS ELAPSED 67
PEOPLE INFECTED 100
DEATHS 70



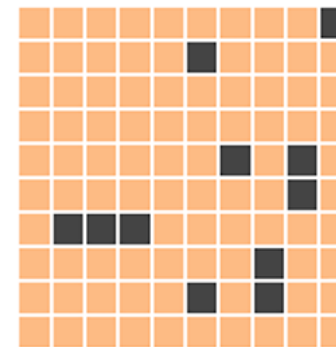
Smallpox ⓘ
DAYS ELAPSED 43
PEOPLE INFECTED 100
DEATHS 30



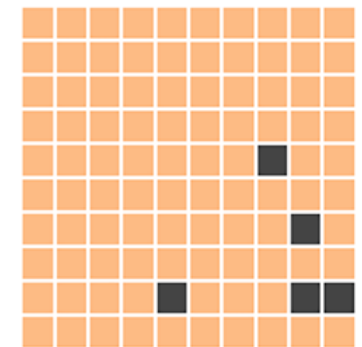
Measles ⓘ
DAYS ELAPSED 23
PEOPLE INFECTED 100
DEATHS 26



SARS ⓘ
DAYS ELAPSED 47
PEOPLE INFECTED 100
DEATHS 11



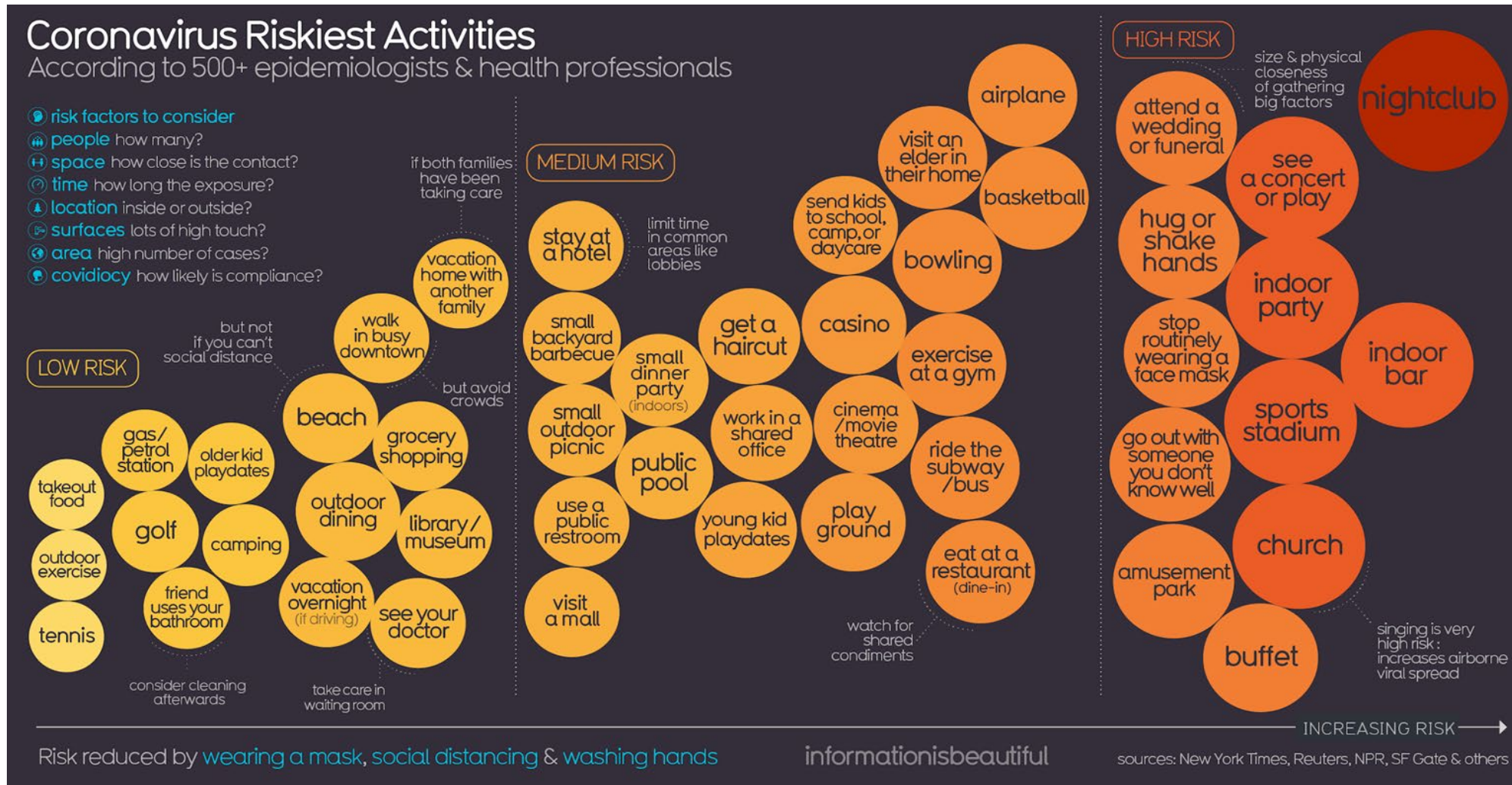
Diphtheria ⓘ
DAYS ELAPSED 77
PEOPLE INFECTED 43
DEATHS 0



<http://www.washingtonpost.com/wp-srv/special/health/how-ebola-spreads/#b10g15t20w14>



Explanatory



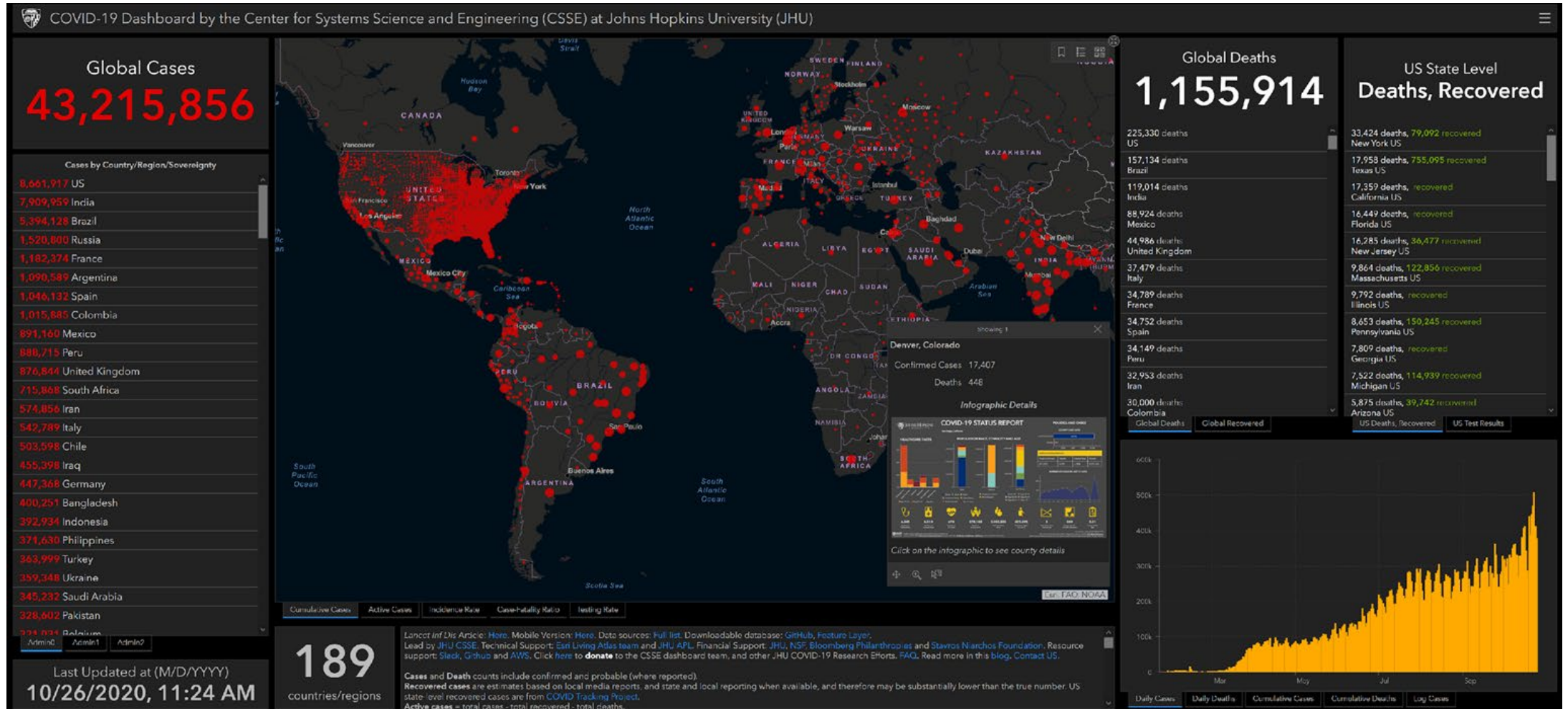


Every Noise: Audio samples of every music genre plotted on a massive scatterplot display



<http://everynoise.com/engenremap.html>

Exploratory



Johns Hopkins University COVID-19 Dashboard: <https://coronavirus.jhu.edu/map.html>



**Story: Consider Gaps
and Begin Analysis**

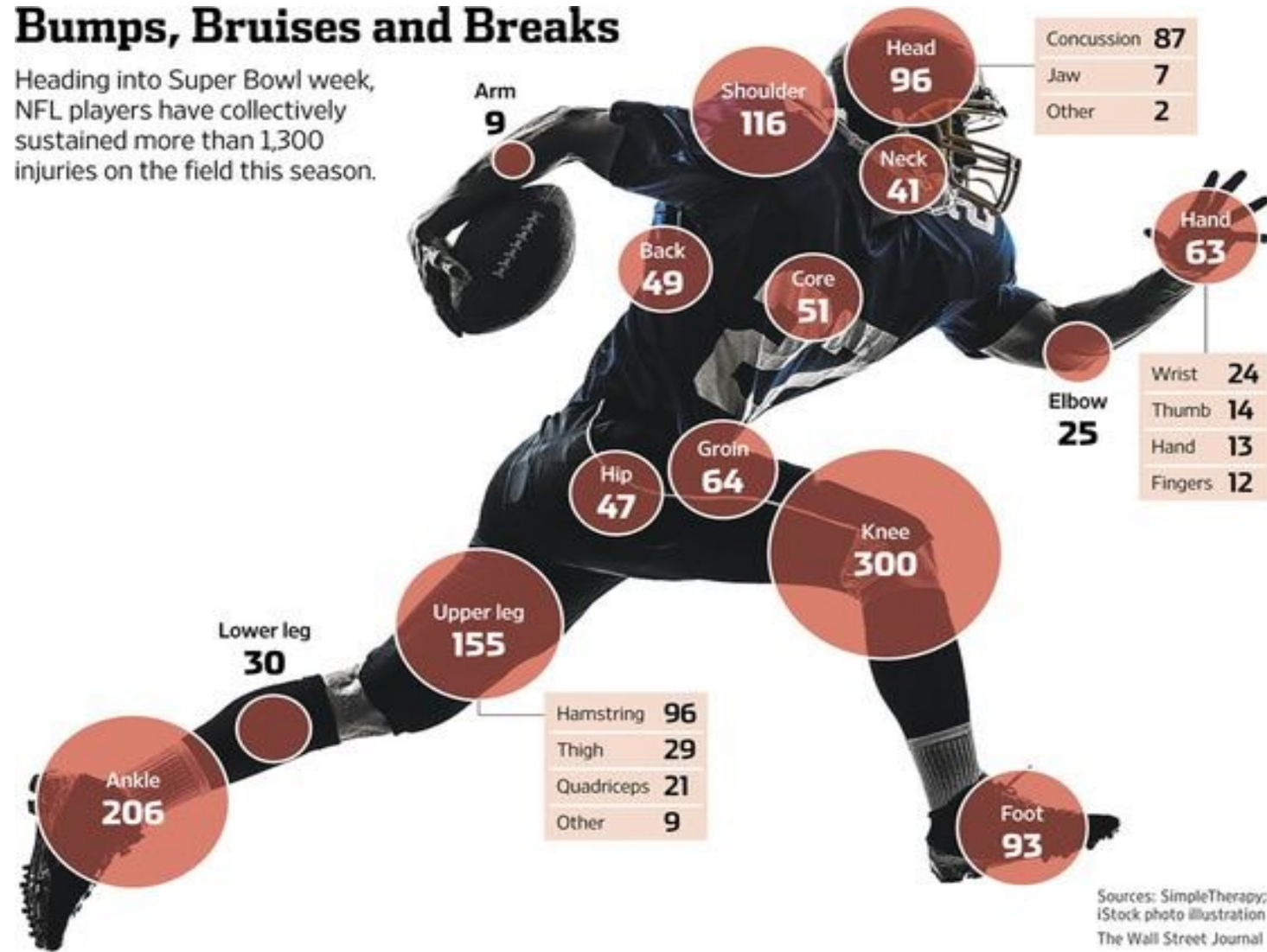
**Find any additional
data you may not have
to tell your story**

**Then begin to organize,
extract, and rearrange**

<https://www.flickr.com/photos/joannaorf/33897259004>

Bumps, Bruises and Breaks

Heading into Super Bowl week, NFL players have collectively sustained more than 1,300 injuries on the field this season.





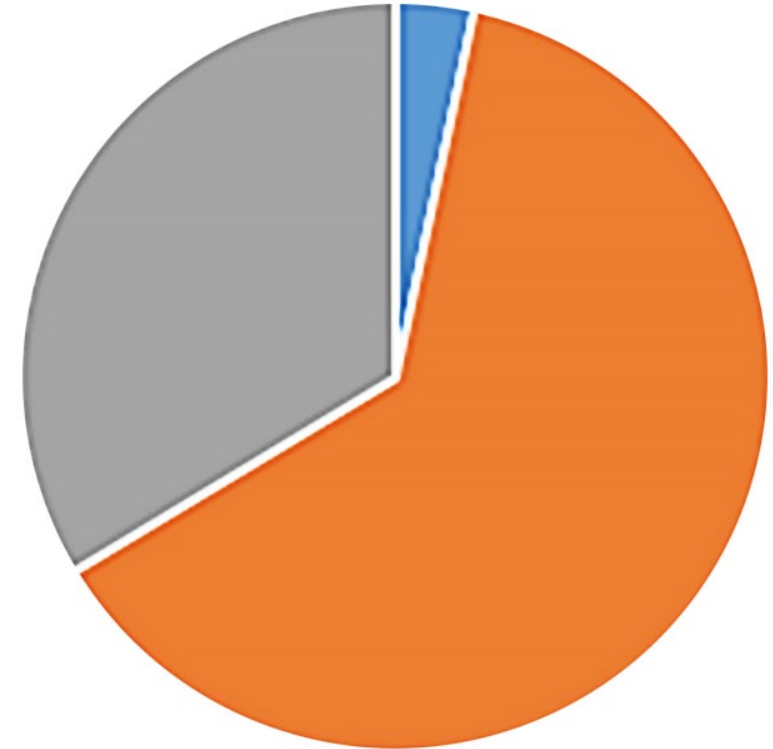
Common Types of Visualizations

Composition

Use to show **parts of a whole**.

- **Pie Charts**
- Bar / Column Charts
- Area Charts

Wildlife Population in 2017



■ Bears ■ Dolphins ■ Whales

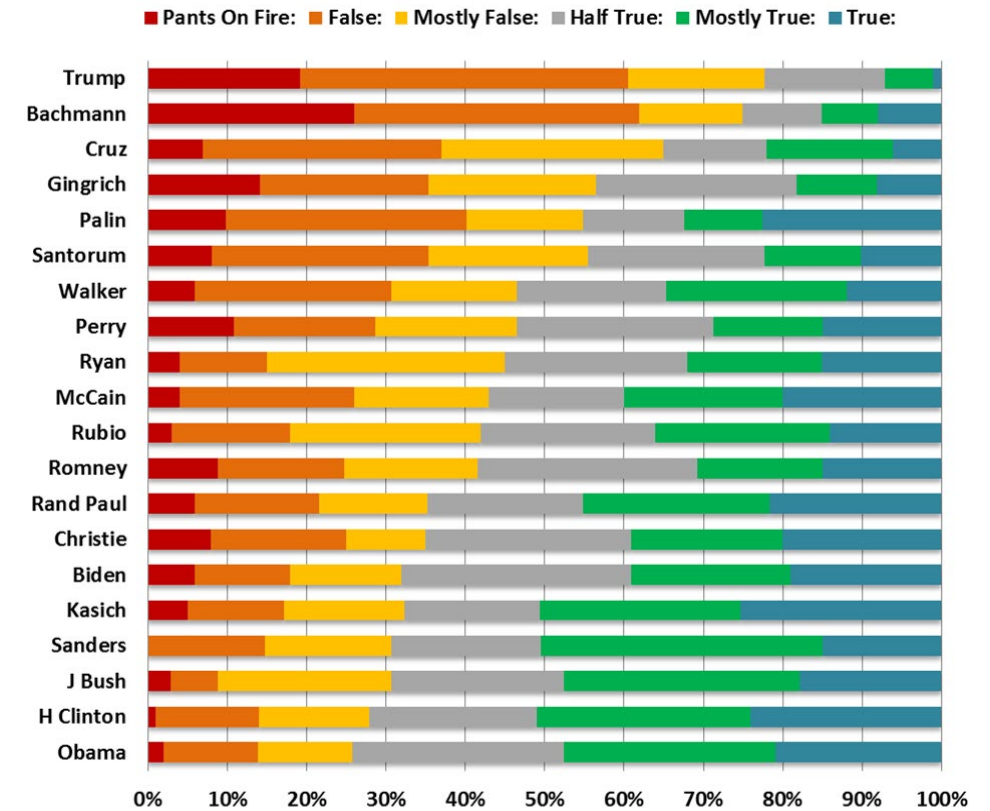
Composition

Use to show parts of a whole.

- Pie Charts
- Bar / Column Charts
- Area Charts

Who Lies More: A Comparison

PolitiFact, an independent fact-checking website, has graded more than 50 statements since 2007 from each of these candidates. Here is how they rank.



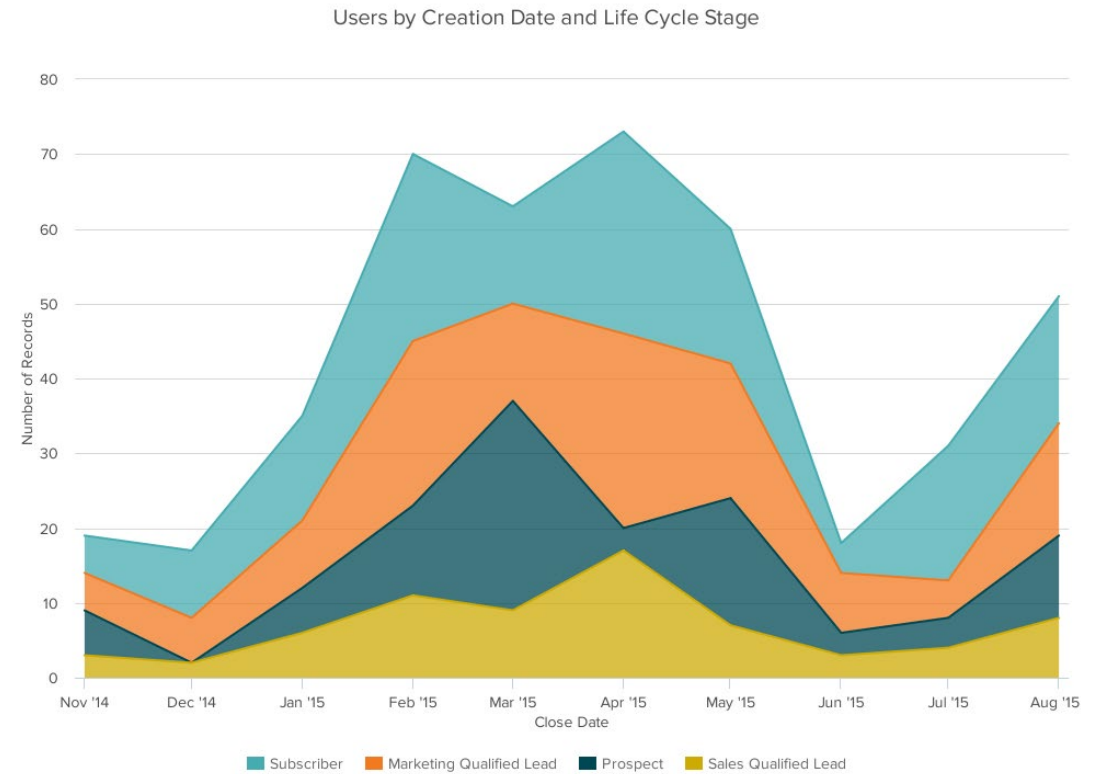
© Robert Mann



Composition

Use to show **parts of a whole**.

- Pie Charts
- Bar / Column Charts
- **Area Charts**



DataHero

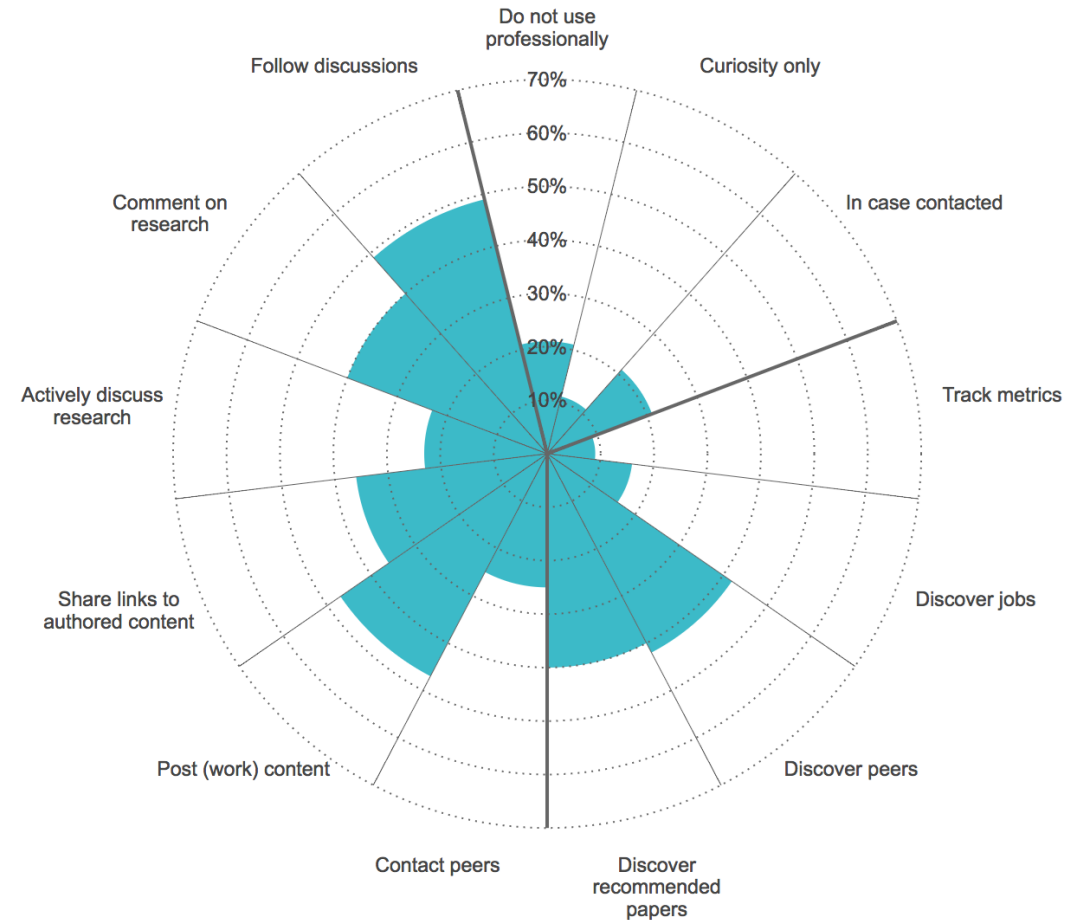




Comparative

Use for comparing, ranking, contrasting data.

- **Circular Area Charts**
- Bubble Charts



Van Noorden, 2014: Online collaboration: Scientists and the social network

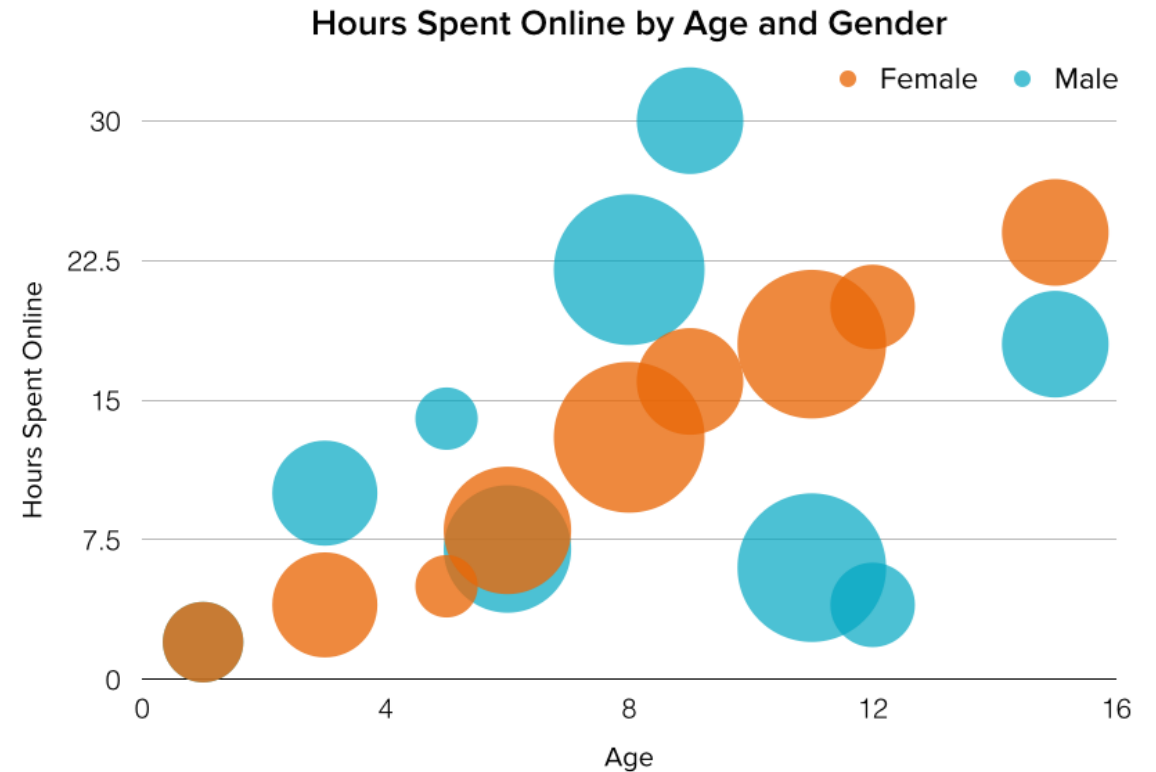




Comparative

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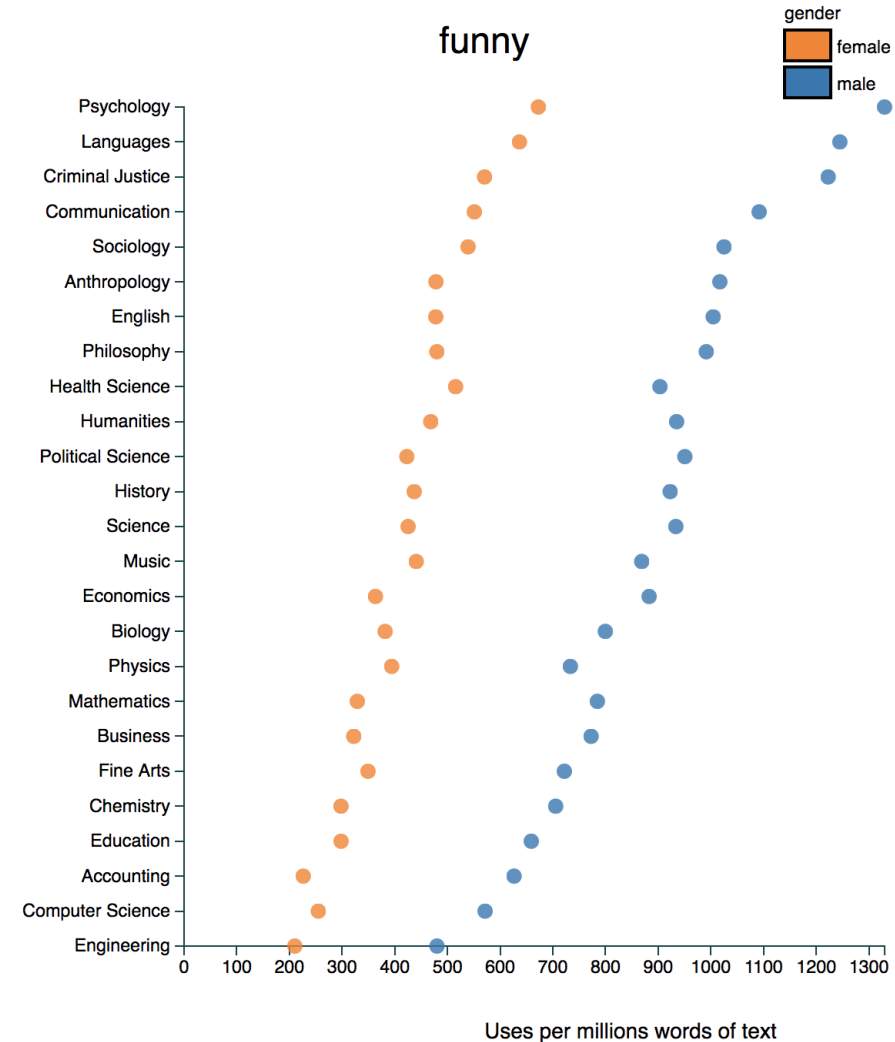




Distribution

Use to visualize correlations and interactions between values.

- **Scatter Plots**
- Histograms
- Alluvial/Sankey Diagrams
- Geospatial
- Heat Maps

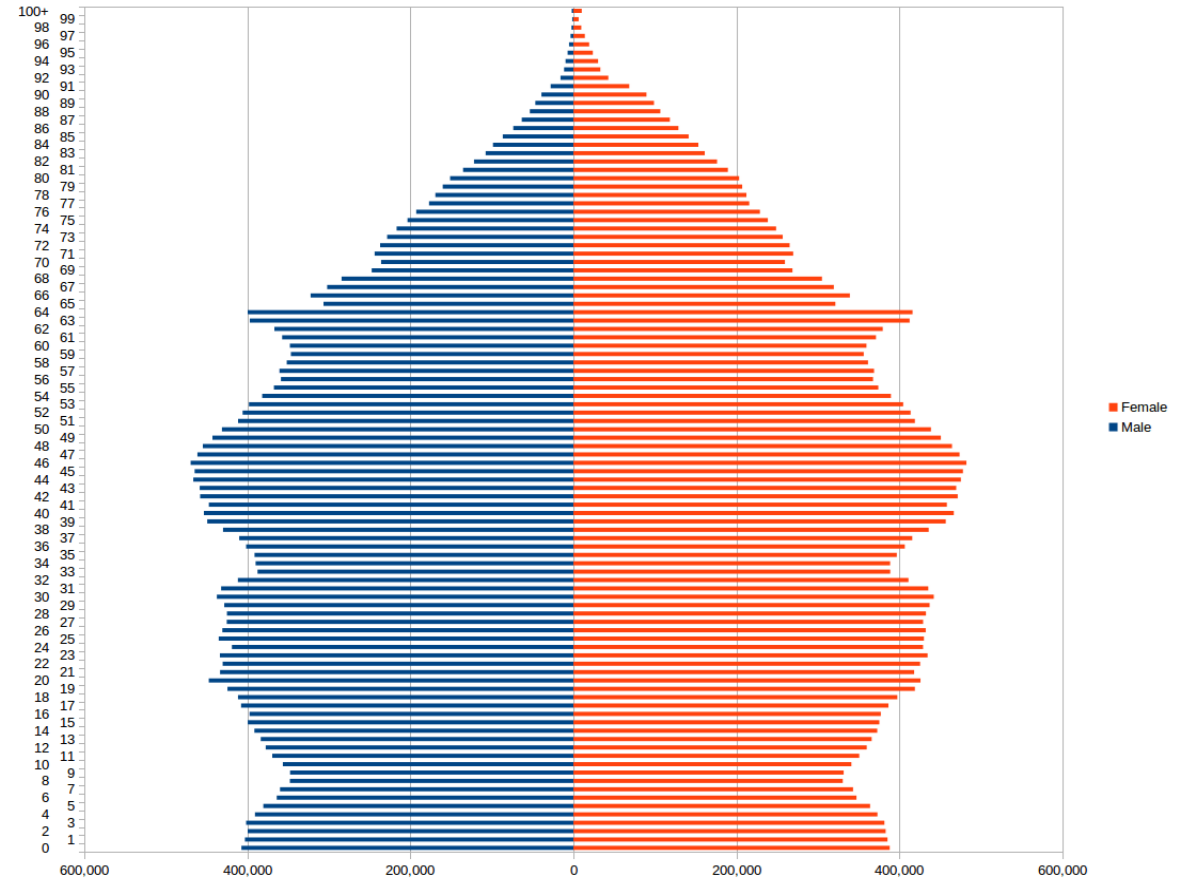




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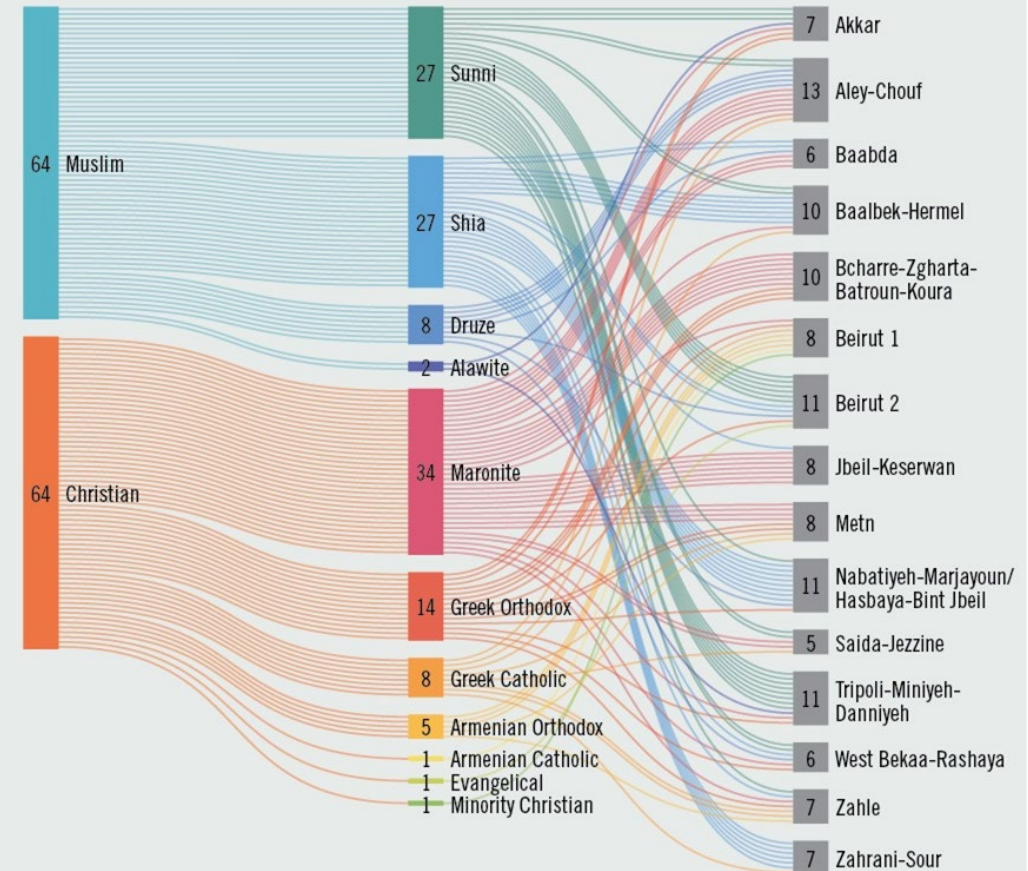
Distribution

Use to visualize correlations and interactions between values.

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- Heat Maps

2017 LEBANESE ELECTION LAW

2018 SEATS BY CONFESSION
Distribution of seats by confession and district



Source: 2017 Election Law; Ministry of Interior and Municipalities.

© 2017 Executive.com.lb



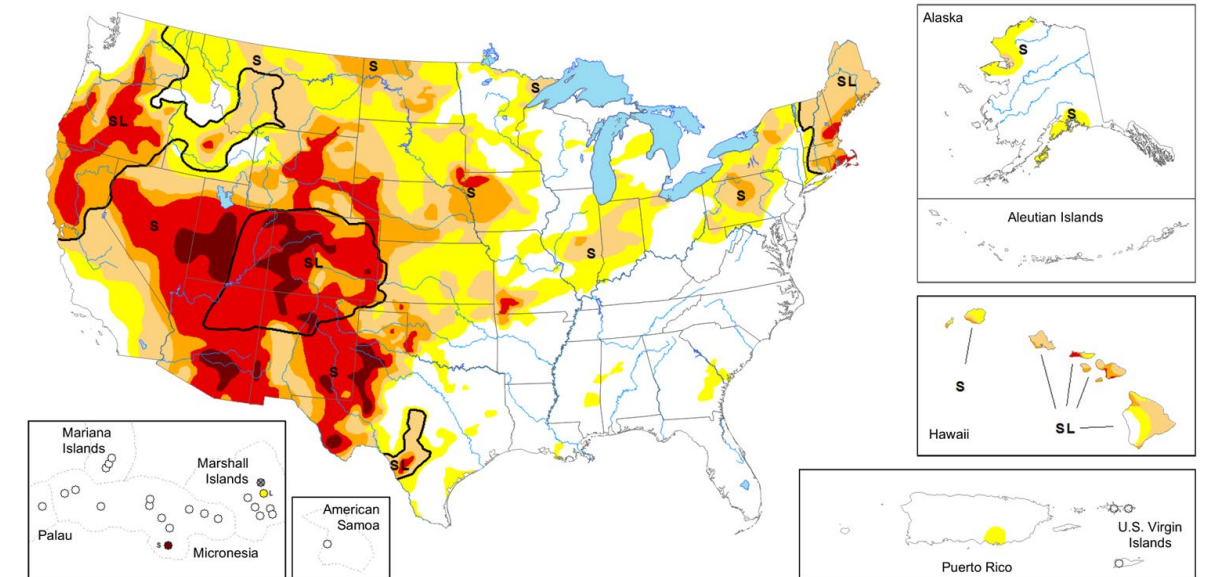
Distribution

Use to visualize correlations and interactions between values.

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- **Geospatial**
- **Heat Maps**

Map released: October 22, 2020

Data valid: October 20, 2020



United States and Puerto Rico Author(s):
Curtis Riganti, National Drought Mitigation Center
Intensity and Impacts



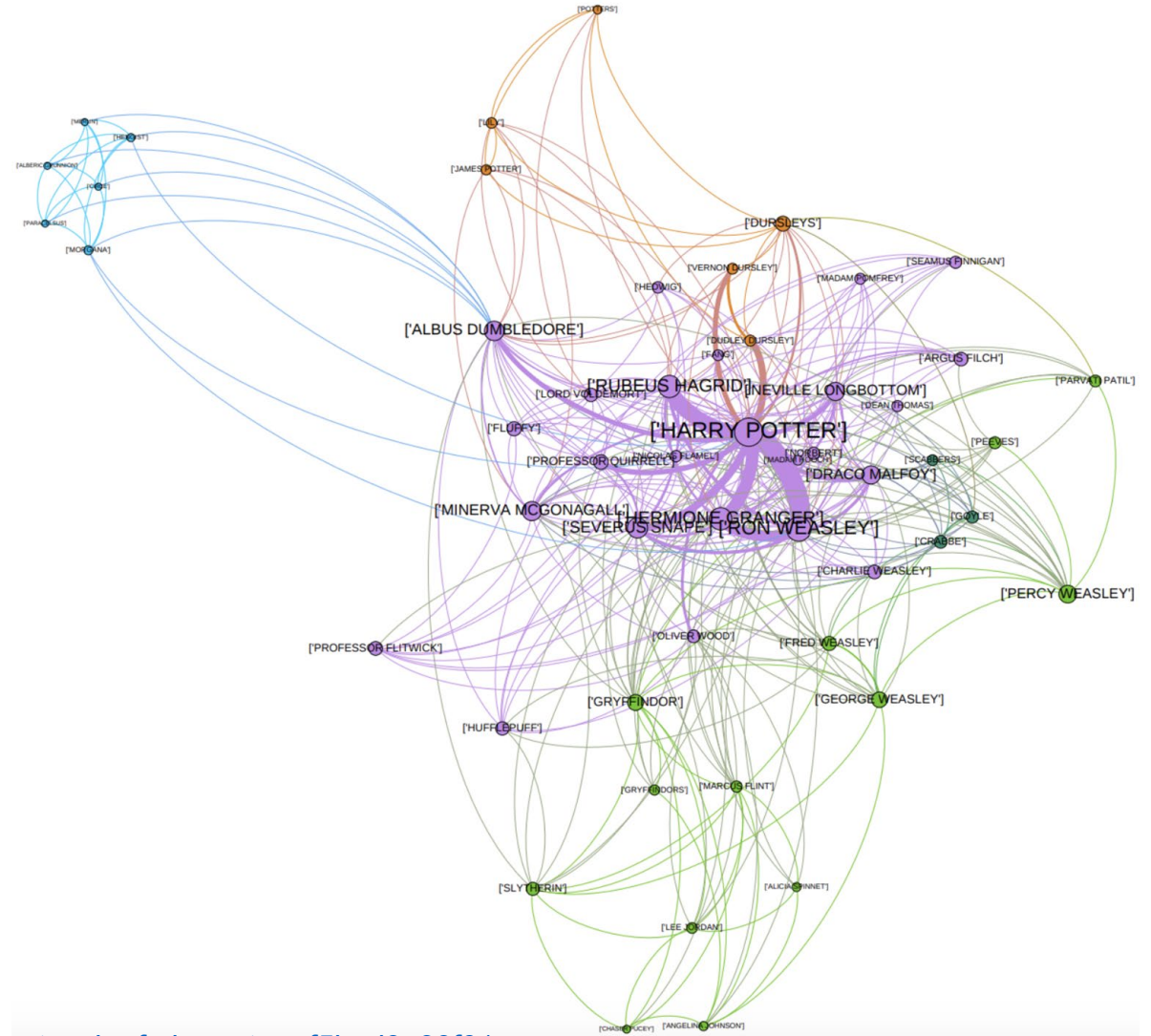
U.S. Affiliated Pacific Islands and Virgin Islands Author(s):
Richard Heim, NOAA/NCEI

~ - Delineates dominant impacts
S - Short-Term impacts, typically less than 6 months (e.g. agriculture, grasslands)
L - Long-Term impacts, typically greater than 6 months (e.g. hydrology, ecology)

<https://droughtmonitor.unl.edu/>

Use for visualizing and tracing relationships and dependencies.

- **Network Graphs**
- Alluvial/Sankey Diagrams
- Tree Graphs
- Treemaps



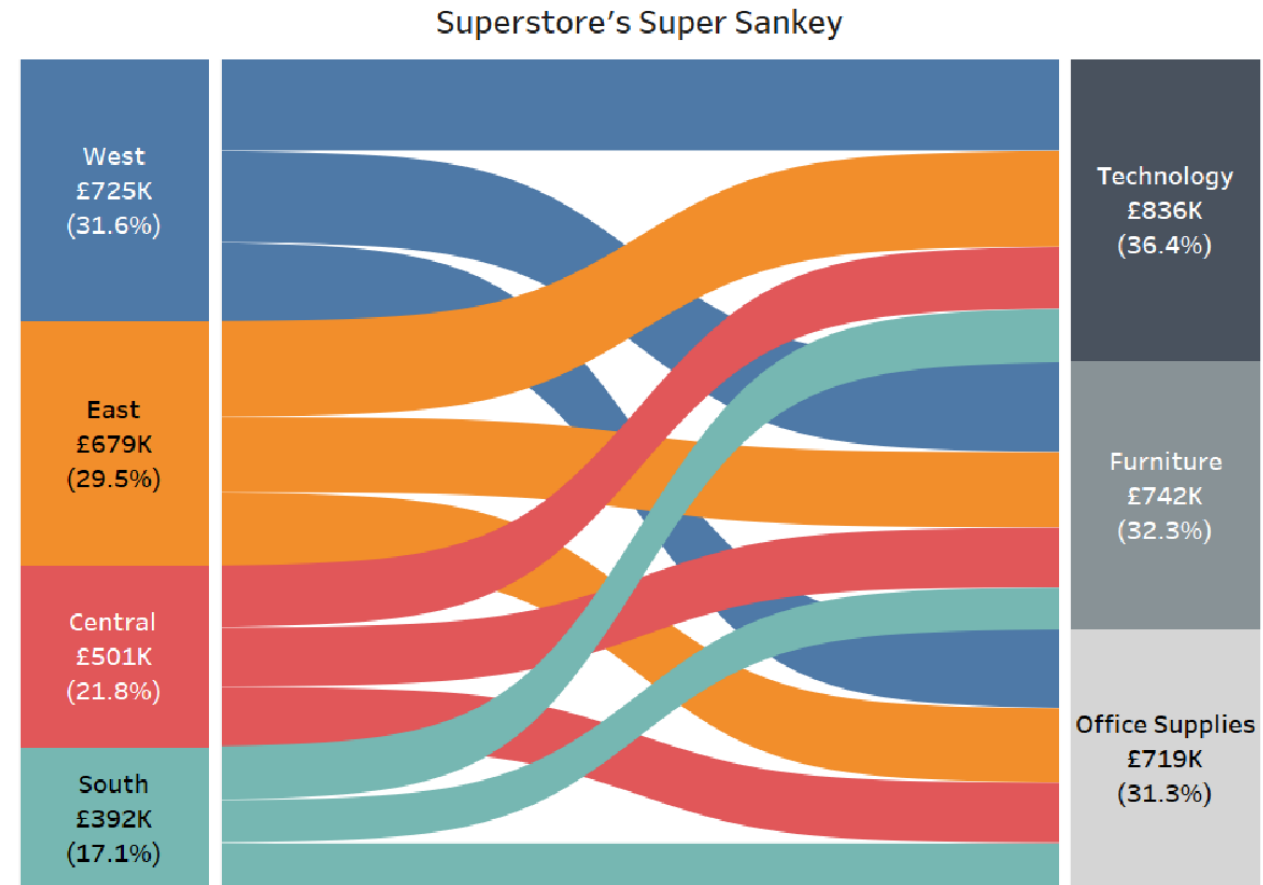
<https://towardsdatascience.com/explore-harry-potter-via-a-dynamic-social-network-of-characters-f5bed9a39f01>



Relational

Use for visualizing and tracing relationships and dependencies.

- Network Graphs
- **Alluvial/Sankey Diagrams**
- Tree Graphs
- Treemaps



<https://www.theinformationlab.co.uk/2018/03/09/build-sankey-diagram-tableau-without-data-prep-beforehand/>

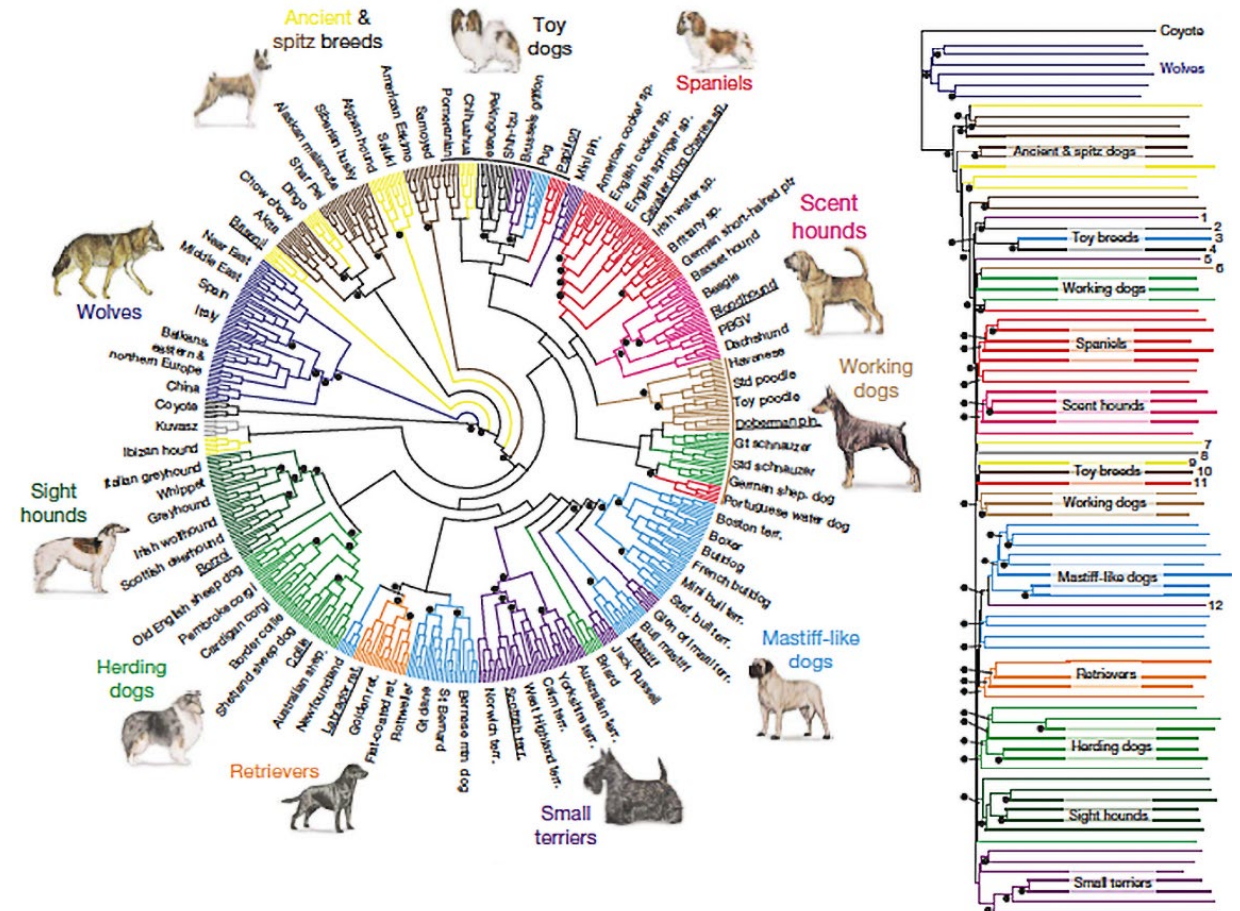




Relational

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<https://www.instituteofcaninebiology.org/how-to-read-a-dendrogram.html>

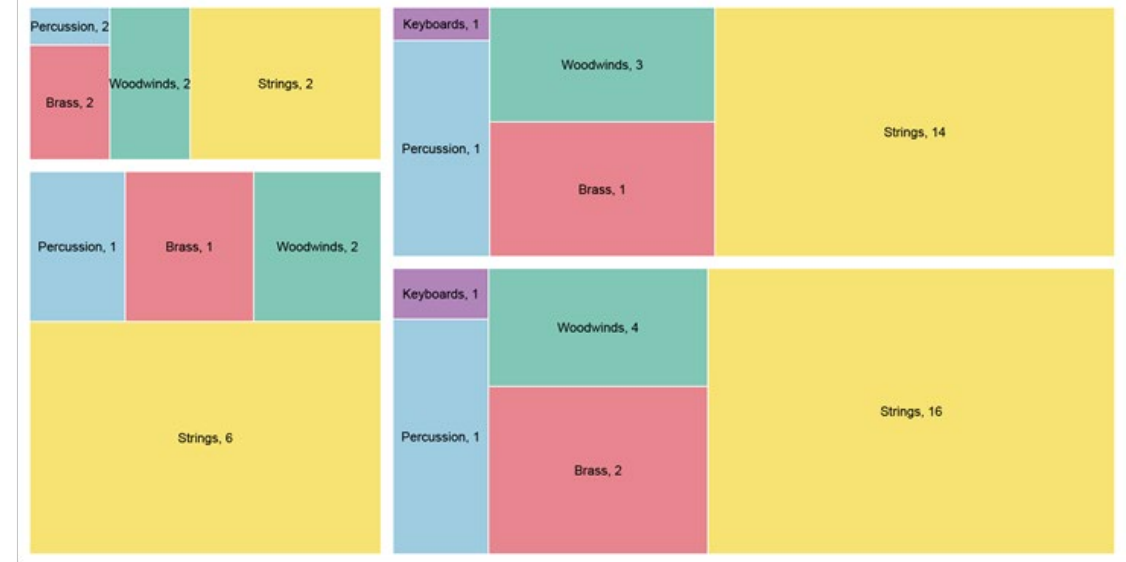




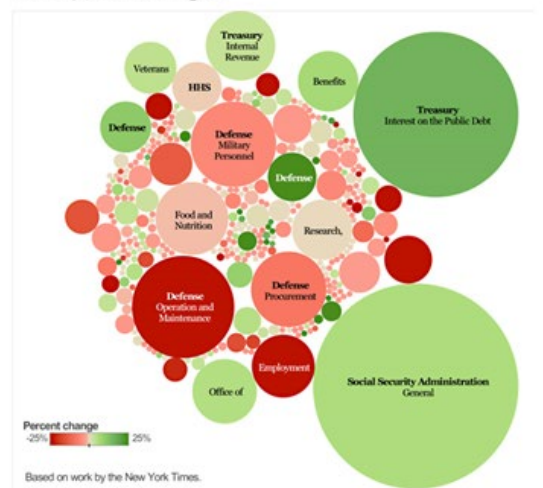
Relational

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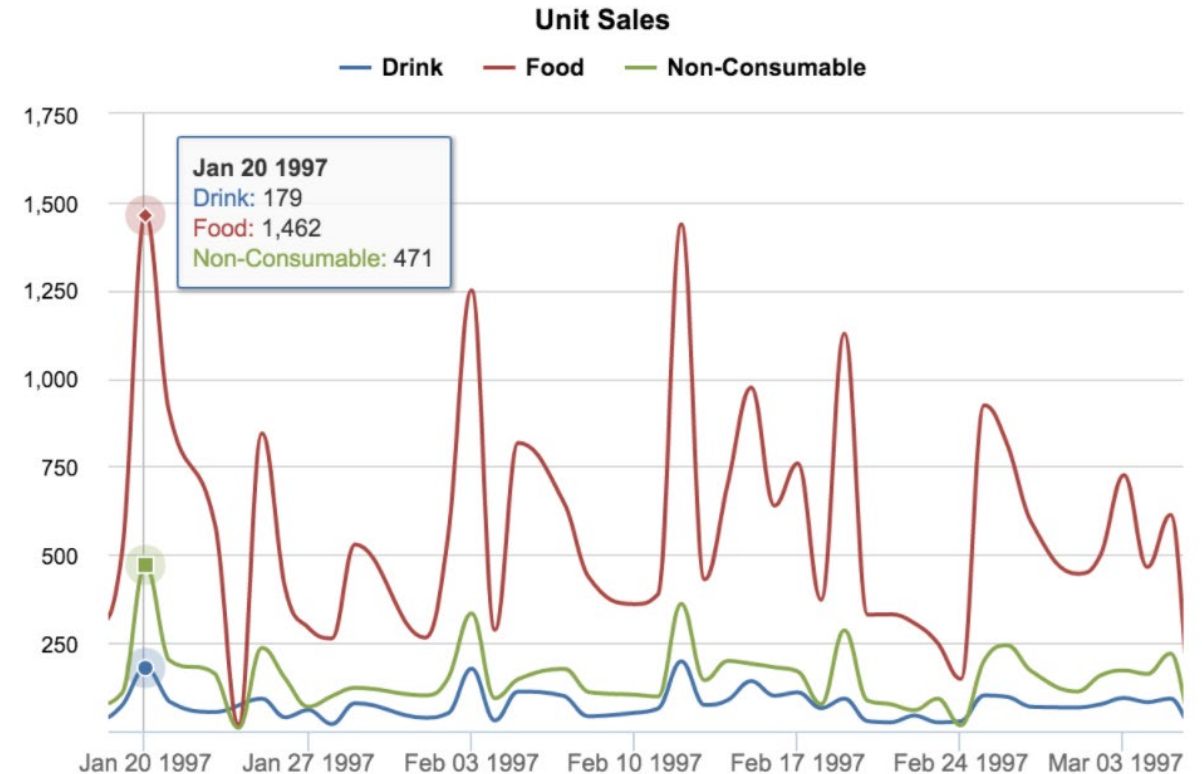


How \$3.7 Trillion is Spent



Use to show change over time.

- **Line, Bar, Column Charts**
- Heat Maps
- Gantt Charts



Use to show change over time.

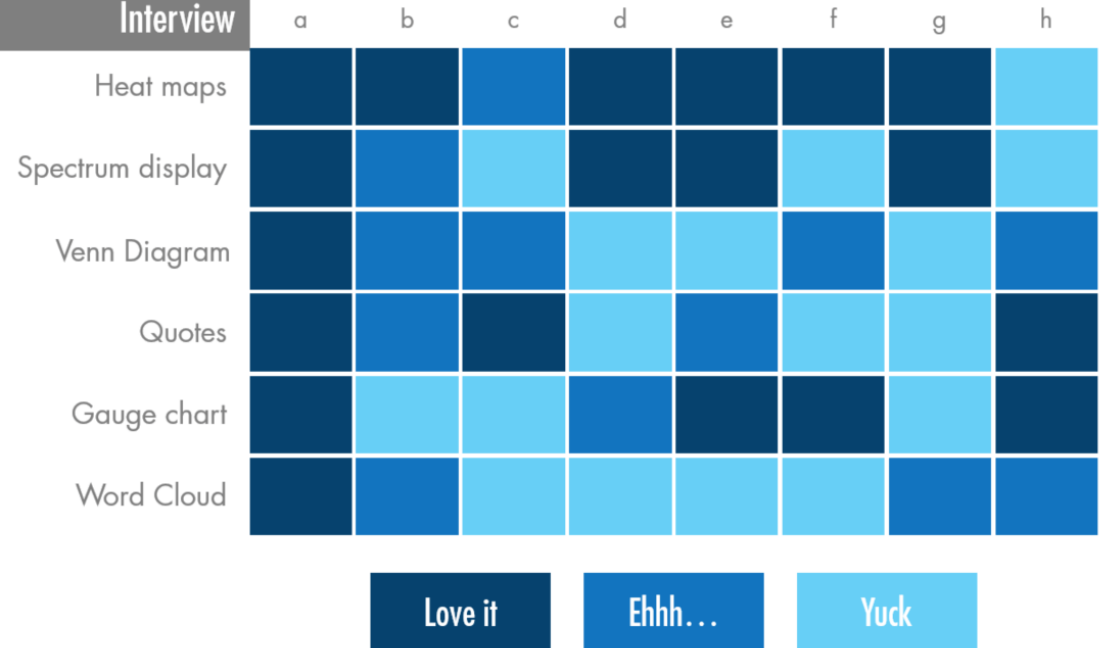
- Line, Bar, Column Charts
- **Heat Maps**
- Gantt Charts

FINDINGS

Qualitative
Visualization
Interview

Most interviewees **loved** heat maps.

Respondents

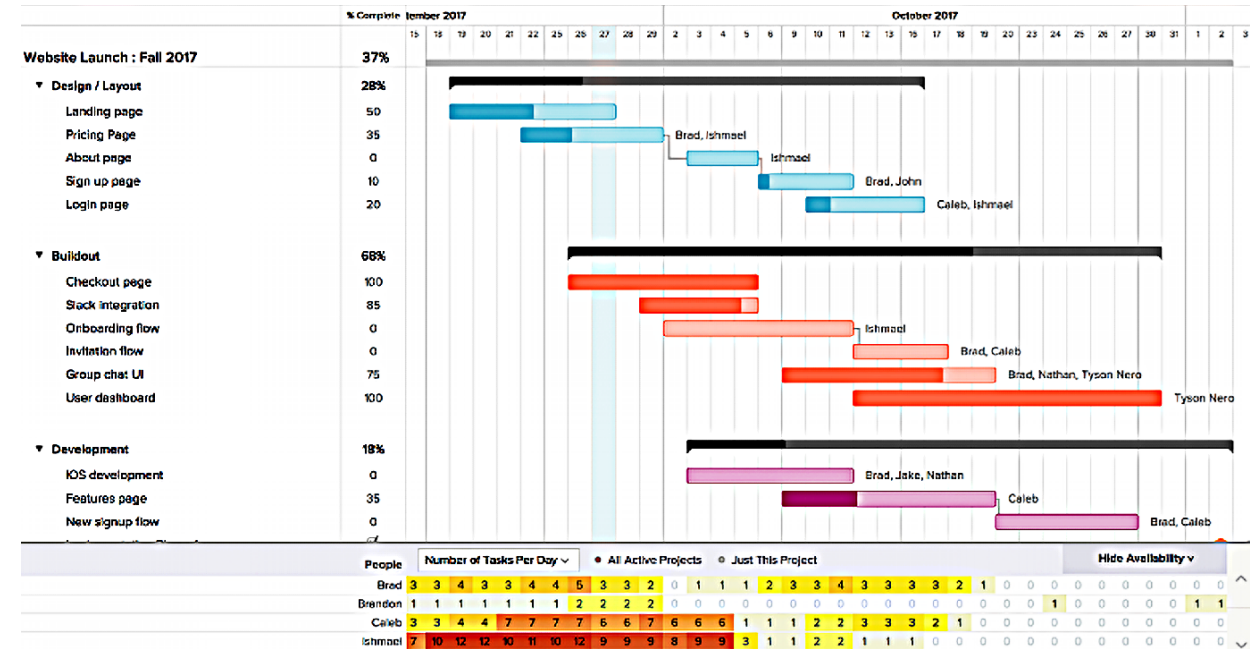


<https://blogs.stockton.edu/datavis/2017/11/20/heat-map-survey/>



Use to show change over time.

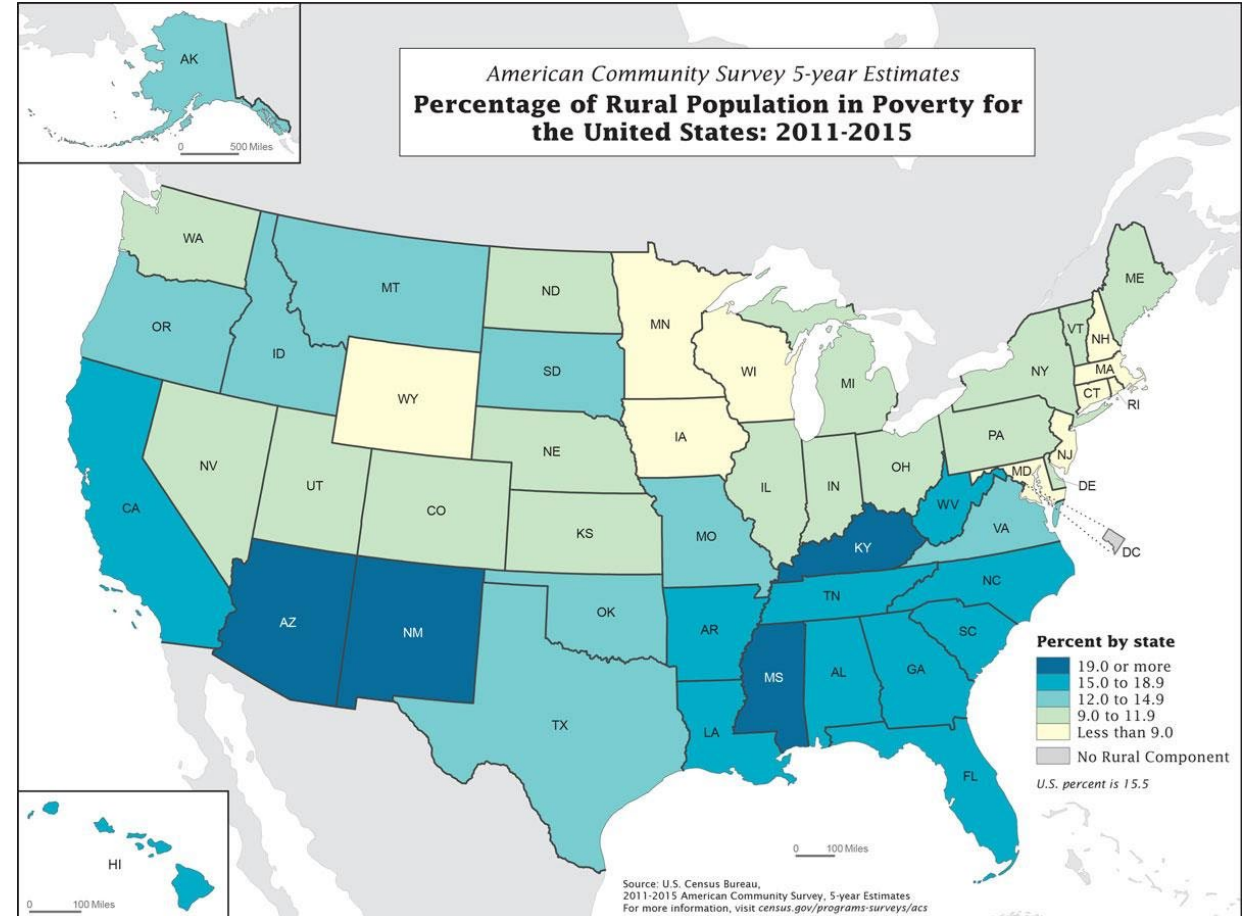
- Line, Bar, Column Charts
- Heat Maps
- **Gantt Charts**



<https://www.teamgantt.com/what-is-a-gantt-chart>

Use to illustrate spatial relationships and distributions.

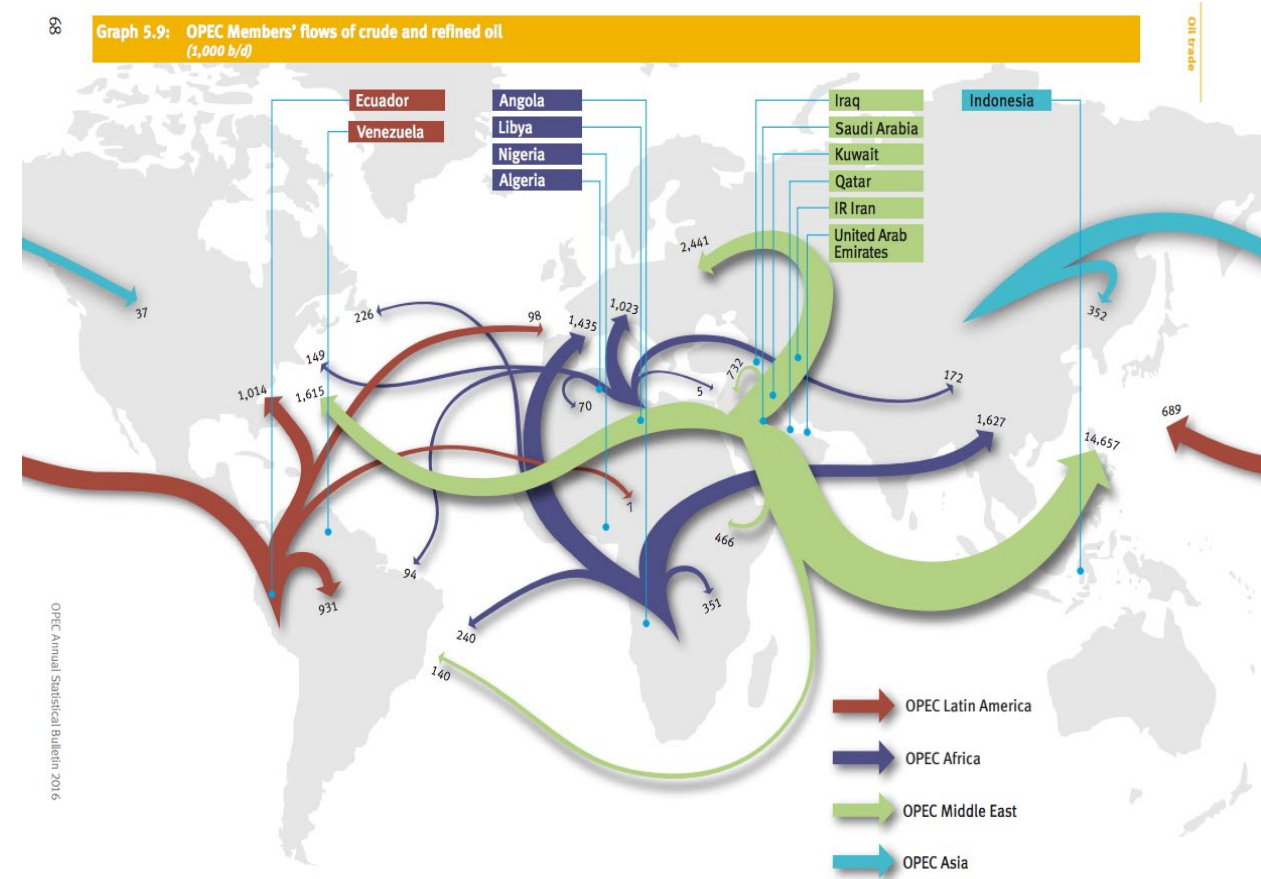
- **Choropleth**
- Flow Maps
- Dot and Bubble Maps
- Network Maps



<https://xdgov.github.io/data-design-standards/visualizations/choropleth-map>

Use to illustrate spatial relationships and distributions.

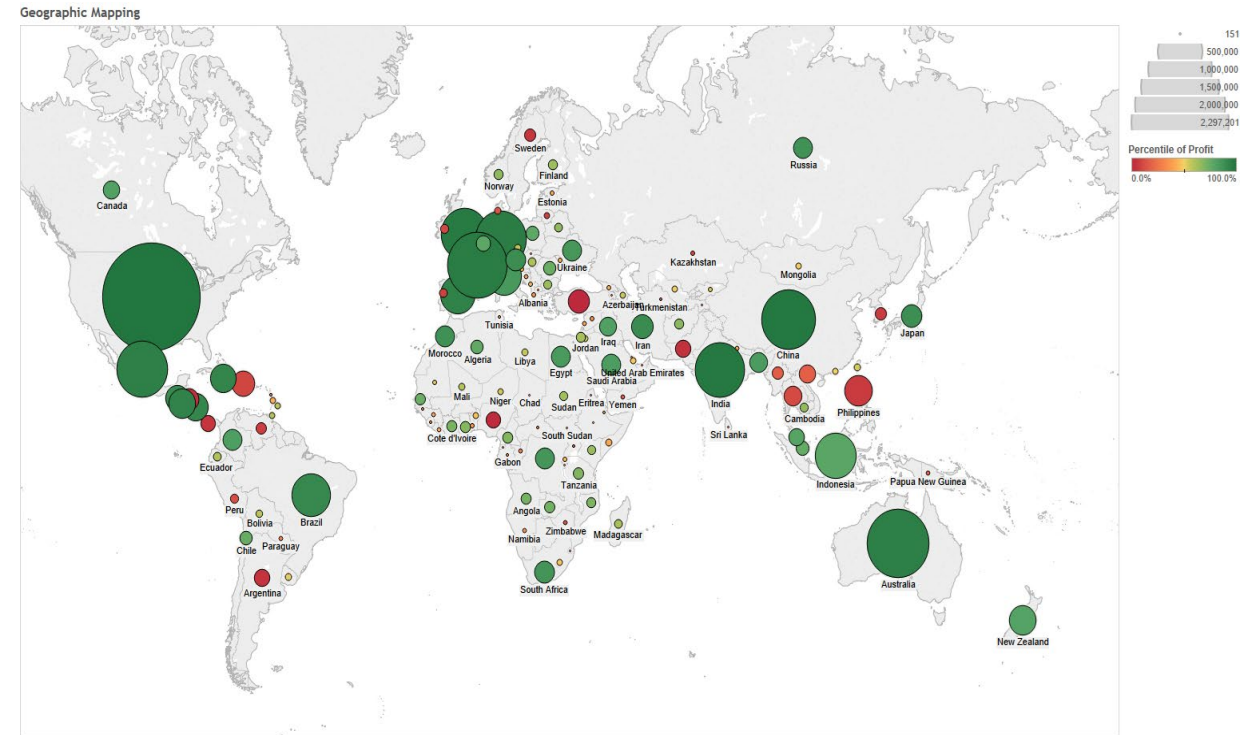
- Choropleth
- **Flow Maps**
- Dot and Bubble Maps
- Network Maps



<https://sanghacpa.com/this-fascinating-map-shows-how-opecs-oil-flows-around-the-world/>

Use to illustrate spatial relationships and distributions.

- Choropleth
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- **Dot and Bubble Maps**
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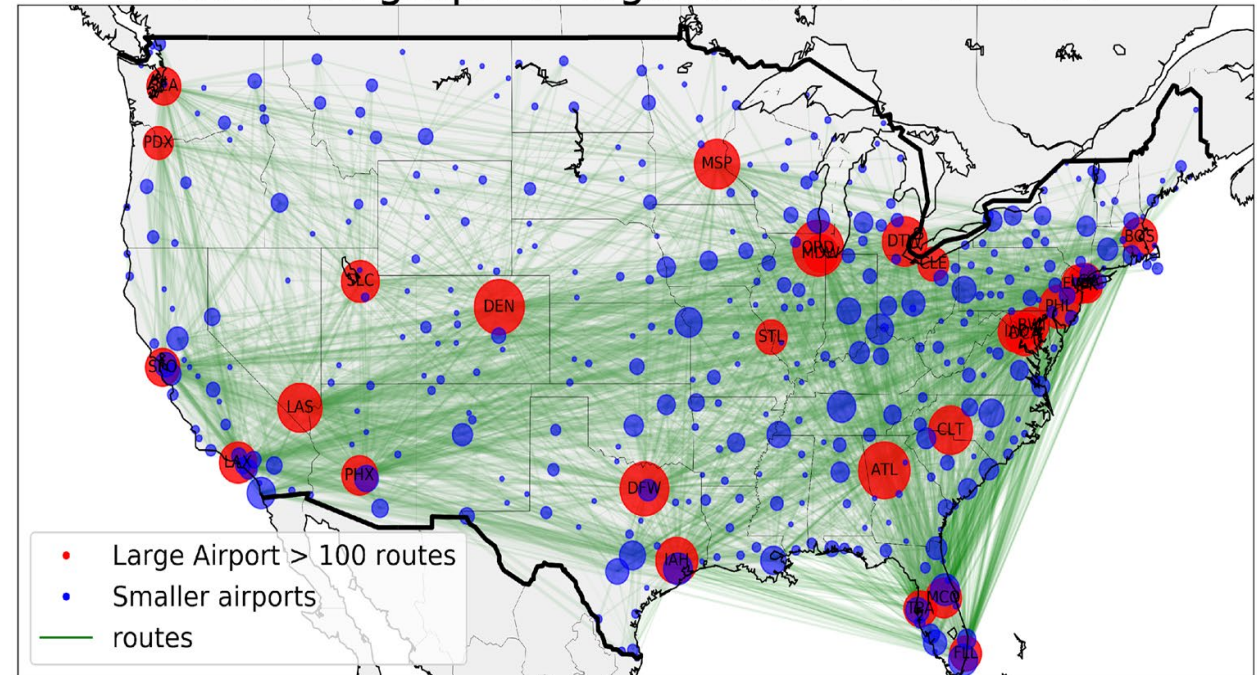


<https://community.tableau.com/s/idea/0874T000000HAgfQAG/detail>

Use to illustrate spatial relationships and distributions.

- Choropleth
- Flow Maps
- Dot and Bubble Maps
- **Network Maps**

Network graph of flight routes in the USA



https://tuangauss.github.io/projects/networkx_basemap/networkx_basemap.html

[illegible]

Be Boulder.



Best Practices



Know your audience

Data visualization can be used in various different media for different expressions of your story.

- Classic Print
- Static web pages
- Interactive web pages
- Animations/movies
- Virtual (eg VR, AR, XR)



Data Integrity

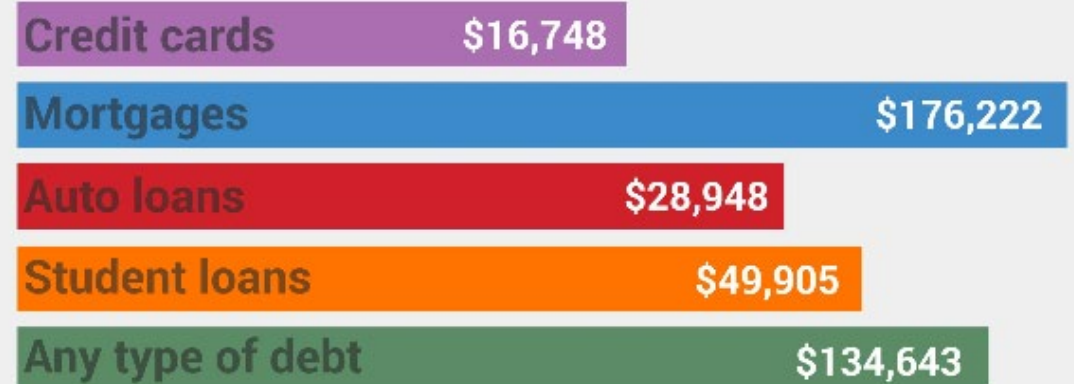


Numeric representation and sorting

- Sort values in ascending or descending order, not alphabetically or by importance.
- With percentages, ensure segments add up to 100.
- Numbers should be directly proportional to the numerical quantities presented.
- Provide correct symbolic units as appropriate

Types of debt

The total owed by the average U.S. household, by debt type.





Legends and labels

- Legends aren't necessary if you have one data category.
- Use labels directly on charts, whenever possible, to avoid indirect look-up.
- If you have to label too many things, your viz is too complex.

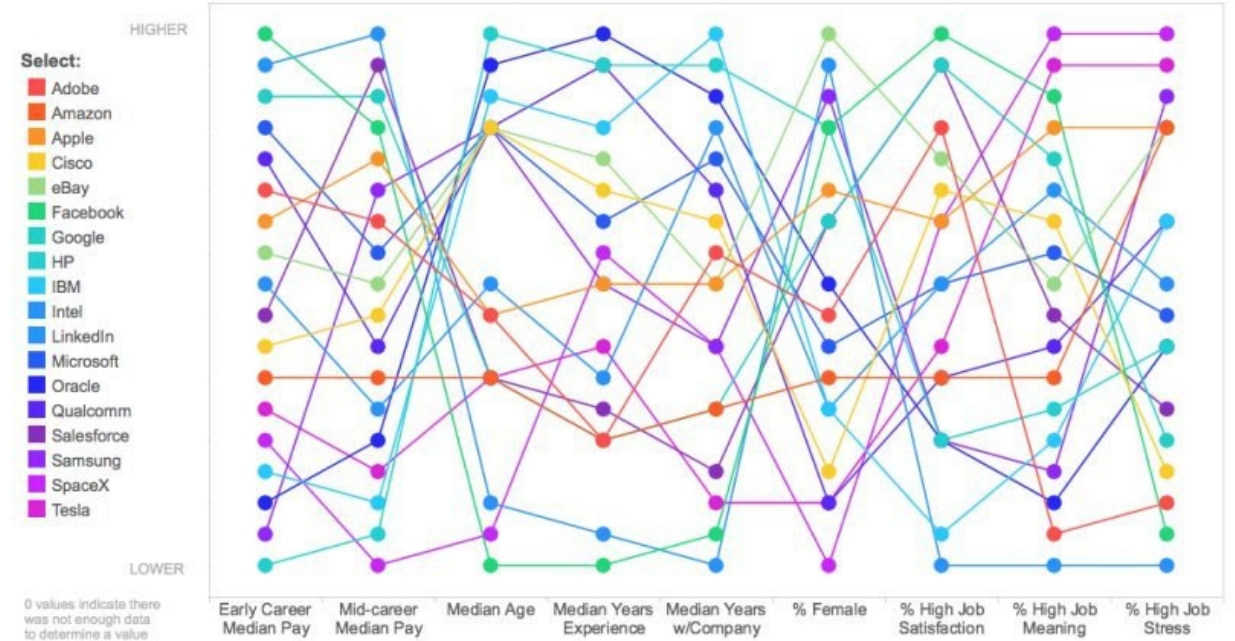
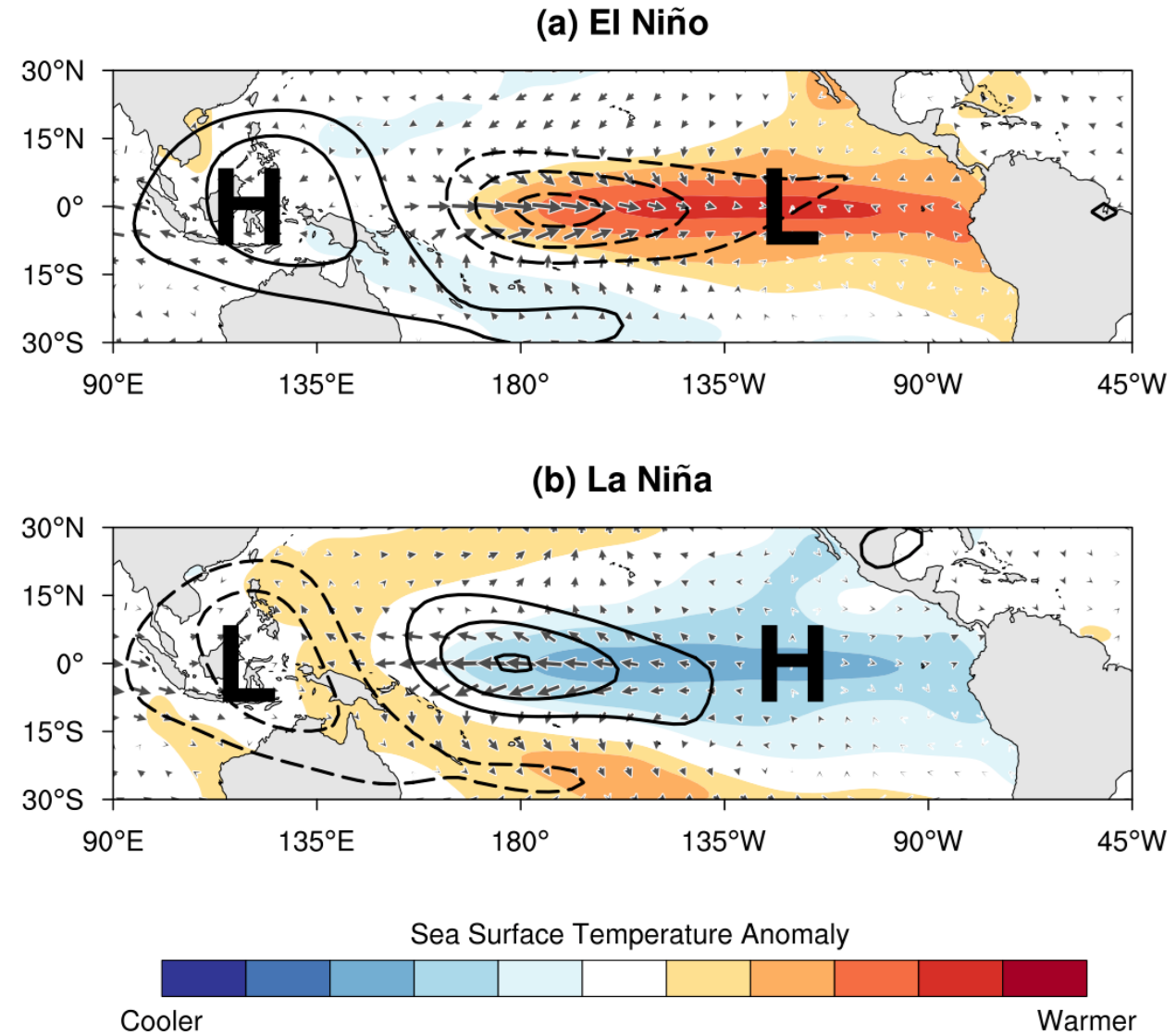


Figure and label sizing

The general of thumb is make your figures, legends, labels, ect, large enough for a person sitting at the back of a large auditorium to clearly read them!





Colors and Patterns

- Use as few colors/patterns as possible
- Only use patterns when necessary
- Verify colors are distinguishable when printed in grayscale.
- Ensure your colors and patterns are accessible for color-blindness.

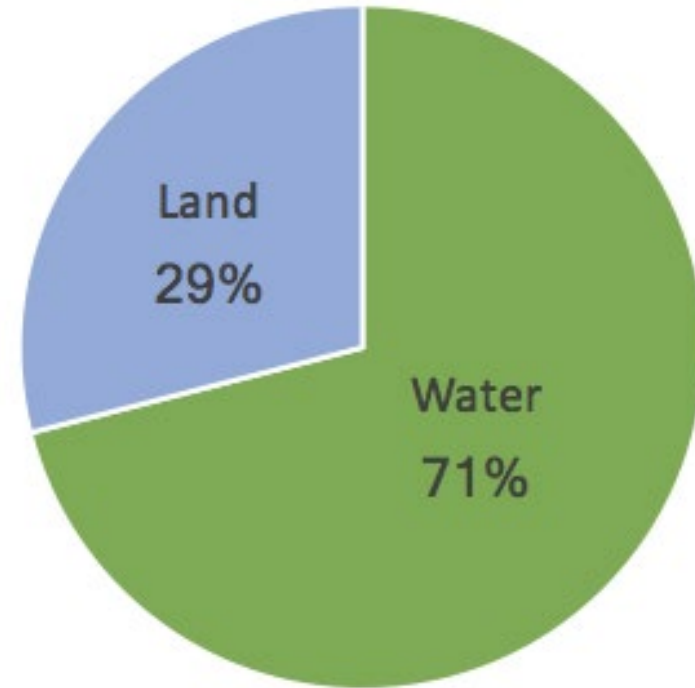




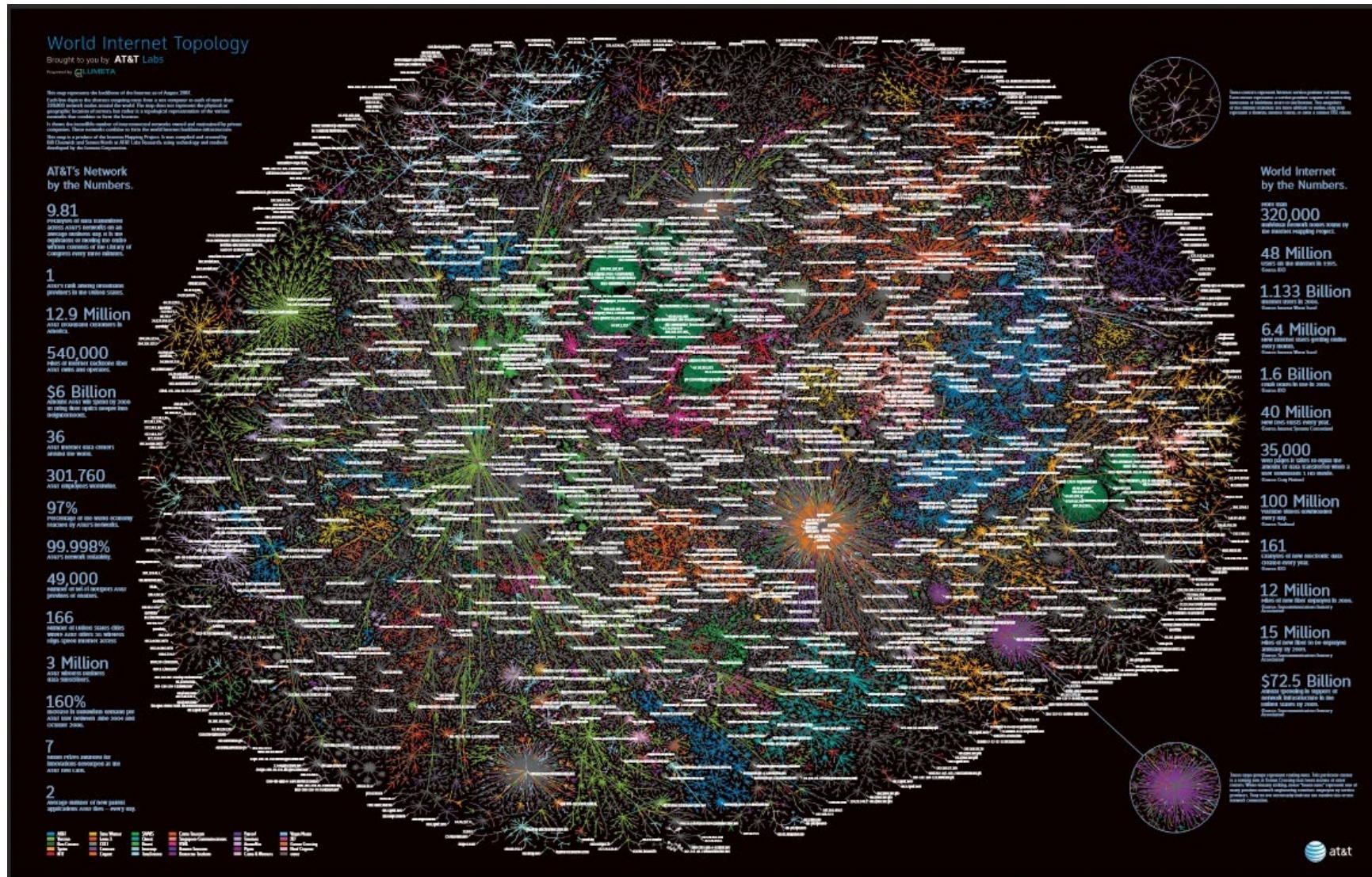
Color, Perception and Reality

- The use of color controls the way a person perceives what's presented.
- Used properly it helps a viewer understand images faster and easier.
- Used poorly, even if only subconsciously, it becomes confusing and may even, albeit unintentionally, lie about the data.

Land to Water Ratio on Planet Earth



The Golden Rule – Keep it SIMPLE!



<https://bourquemaps.wordpress.com/>

CRDDS

Open Consulting Hours

Tuesday 12-1pm

Thursday 1 - 2pm

or by appointment

<https://www.colorado.edu/crdds/>





Thank You!

Questions?